

STA305: taught, drilled, and a predicted paper

Do each question on paper first, then read the solution below it. Every step says **why** that number is there. Highlight freely.

ONE SCREEN. EVERYTHING NOT ON THE FORMULA SHEET.

Reference card

DESIGN	NAMED ROWS AND DF	ERROR DF	TOTAL
One-way	Treat $a - 1$	$N - a$	$N - 1$
Two-way	A $a - 1$, B $b - 1$, AB $(a - 1)(b - 1)$	$N - ab$	$N - 1$
RCB	Treat $a - 1$, Block $b - 1$	$N - a - b + 1$	$N - 1$
Latin square	Treat, Row, Col each $a - 1$	$(a - 1)(a - 2)$	$a^2 - 1$
BIBD	Treat $a - 1$, Block $b - 1$	$N - a - b + 1$	$N - 1$
ANCOVA	Covariate 1, Treat $a - 1$	$N - a - 1$	$N - 1$

THING	FORMULA OR FACT
Grand mean	$\bar{Y}_T = \frac{1}{N} \sum n_j \bar{Y}_j$. Plain average only if balanced.
SSE from summaries	$\sum (n_j - 1) s_j^2$. If given sd , square it first.
MSE	$SSE / (N - a)$, and it is $\hat{\sigma}^2$, and it is the denominator of every F.
4 design principles	Control, Block, Randomize, Replicate , in that order.
Orthogonal	$\sum c_{1j} c_{2j} / n_j = 0$; balanced gives $\sum c_{1j} c_{2j} = 0$
Contrast	$SS_\psi = \hat{\psi}^2 / \sum (c_j^2 / n_j)$, 1 df, $F \sim F_{1, N-a}$
Bonferroni	$\alpha_B = \alpha / k$
Sidak-Bonferroni	$\alpha_{SB} = 1 - (1 - \alpha)^{1/k}$
Family size	all pairwise: $k = \binom{a}{2}$ (a=4 gives 6, a=5 gives 10)
Post-hoc choice	all pairwise $\rightarrow$ Tukey; planned contrasts $\rightarrow$ Bonferroni/Sidak; unplanned $\rightarrow$ Scheffe
Dummy coding	$\beta_0 = \mu_{\text{ref}}$; $\beta_p = \mu_p - \mu_{\text{ref}}$
Effect coding	$\alpha_0 = \bar{Y}_T$; $\alpha_p = \mu_p - \bar{Y}_T$
Cell means	no intercept; $\gamma_p = \mu_p$; $H_0 : \gamma_1 = \dots = \gamma_a$
Kruskal-Wallis	$H = \frac{12}{N(N+1)} \sum \frac{R_j^2}{n_j} - 3(N+1)$, reject if $> \chi_{\alpha, a-1}^2$
Cohen	small .01/.2, medium .06/.5, large .15/.8 ($R^2 / d $)
BIBD	$N = ar = bk$, $\lambda = \frac{r(k-1)}{a-1}$ must be an integer, $Q_i = T_i - B_i/k$ sums to 0
Proof words	write by independence , and divide by σ^2 . His key deducts a mark otherwise.

PART 0 • READ ONCE, EVERYTHING LATER ASSUMES IT

What are we actually doing, and what does every symbol mean

The one question this whole course asks

You have several groups. Their averages are different. **But averages are always a bit different, just by luck.** So the only real question is:

Is the gap between the group averages bigger than the random noise inside the groups?

That is it. Twelve units, one question. Everything else is bookkeeping about *what counts as noise*.

Why it is called Analysis of Variance when you are comparing means

You compare means by **comparing two kinds of variance**.

WHY Take the spread **inside** each group: that must be pure noise, because everyone in a group got the identical treatment. Now take the spread **between** the group averages. If the treatment does nothing, that between-spread is also just noise, and the two numbers come out about the same. If the treatment does something, only the between-spread grows. So you divide one by the other and see if the ratio is near 1 or much bigger. That ratio is F .

Why not just do lots of t tests?

With $a = 5$ groups there are 10 pairs. Ten tests at 5% each gives about a **40%** chance of at least one false alarm.

WHY That is the **multiple comparisons problem**, and it is why the course goes: one global F test first (is *anything* going on?), then, only if that rejects, a controlled follow-up (contrasts, Tukey, Bonferroni) to find out *where*. Units 3 and 4 are exactly these two halves.

Every symbol, with what it actually means

SYMBOL	NAME	WHAT IT MEANS IN PLAIN WORDS	EXAMPLE
a	number of groups	How many different treatments you are comparing. Count the rows of the summary table.	3 pesticides $\rightarrow a = 3$
j	group index	A label saying "which group". Runs 1 to a .	$j = 2$ means group 2
i	observation index	"Which person inside that group".	$i = 4$, 4th subject
n_j	group size	How many observations are in group j. The little j is there because groups can be different sizes.	$n_2 = 6$: six farms got pesticide 2
N	total size	Everybody, added up: $N = \sum_j n_j$.	$5 + 6 + 4 + 5 = 20$
Y_{ij}	one observation	The actual measured number for person i in group j .	farm 3 on pesticide 2 yielded 3.7 t
$\bar{Y}_j$	group mean	The average of group j on its own. The bar means "average of"; the j means "of group j only".	$\bar{Y}_2 = 3.65$
$\bar{Y}_T$	grand mean	The average of everybody pooled together , ignoring groups. T is for Total.	$\bar{Y}_T = 3.4$
s_j	group sd	How spread out group j is. Roughly the typical distance of a member of group j from its own mean.	$s_2 = 2.3$
s_j^2	group variance	The square of that. Formulas want the variance, so if you are given s_j you must square it first.	$s_2^2 = 5.29$
σ^2	true noise variance	The real, unknown background variability. Assumed the same in every group . You never see it; you estimate it.	estimated by MSE
μ_j	true group mean	The real population mean for treatment j. $\bar{Y}_j$ is your estimate of it.	$H_0: \mu_1 = \mu_2$
α_j	treatment effect	How far treatment j pushes the response away from the overall average: $\alpha_j = \mu_j - \mu$. They always sum to zero.	$\hat{\alpha}_2 = 6.375$
ε_{ij}	error	The leftover wobble for one observation: the bit the model cannot explain.	$\varepsilon \sim N(0, \sigma^2)$

The three sums of squares, in words

SST (Total) = $\sum \sum (Y_{ij} - \bar{Y}_T)^2$
WHY All the variation there is. Take every single observation, measure how far it is from the overall average, square it, add up. Squaring is only so that pluses and minuses cannot cancel.
SSA (Treatment, "between") = $\sum n_j (\bar{Y}_j - \bar{Y}_T)^2$
WHY The part the grouping explains. How far is each <i>group's</i> average from the overall average? The n_j is there because a whole group of n_j people sits at that average, so it counts n_j times.
SSE (Error, "within") = $\sum (n_j - 1) s_j^2$
WHY The part left over: the noise. How far is each observation from <i>its own</i> group's average. Since s_j^2 already measures that spread divided by $n_j - 1$, multiplying back by $n_j - 1$ recovers group j's raw sum of squares.
$SST = SSA + SSE$ always. Total = explained + leftover.

Why divide by df, and what df even is

$MS = SS/df$, and df = how many independent pieces of information went into that sum. Every mean you estimate from the data costs you one.
WHY A sum of squares is bigger just because you added more terms, so you cannot compare SSA to SSE directly. Dividing by df turns each into an average , which makes them comparable. Treatment df is $a - 1$ (a group means, minus 1 spent on the grand mean); error df is $N - a$ (N observations, minus 1 per group for each group mean).
MSE is the single most important number on the page.
WHY It is your estimate of σ^2 , the background noise. It is the denominator of every F test in this entire course , in every design. When you learn a new design, the only thing that really changes is <i>what got taken out of MSE first</i> .

How every later unit is the same idea

UNIT	THE NEW QUESTION	WHAT CHANGES MECHANICALLY
3 One-way	Do the groups differ at all?	The base case
4 Contrasts	<i>Which</i> groups differ, and how do I ask without inflating the error rate?	Split SSA into 1-df questions; shrink alpha
5 Coding	Can I write this as a regression?	Same model, new parametrisation
6 Kruskal-Wallis	What if the assumptions fail?	Use ranks instead of values
7 Power	Is the difference <i>big</i> , not just significant?	Effect size, no new SS
8 Two-way	Two factors: do they interact?	Two more named rows, plus AB
9 RCB	Can I remove a known nuisance?	Block row steals variance from SSE
10 Latin square	Two nuisances at once?	Two more rows steal from SSE
11 BIBD	What if a block is too small to hold everything?	Treatment totals need adjusting
12 ANCOVA	What if the nuisance is a number, not a category?	Covariate row steals from SSE

THE SENTENCE THAT TIES IT TOGETHER

Blocking, Latin squares and covariates all do the **same job**: pull explainable variation out of the error term. Smaller MSE means a bigger F for the treatment, which means **more power to detect a real effect**. That is the entire motivation for Units 9 to 12, and it is worth being able to say in one line.

PAPER A • STA305 TERM TEST 1, FEBRUARY 2025 • 20 MARKS, 75 MIN

Full past paper

A1 [2 MARKS]

A school district is investigating the impact of **device type** (tablet, laptop) and **study app** (App A, App B, App C) on student test scores. Balanced design, 6 replicates per treatment. Participants randomly assigned. Scores recorded as a percentage.

- Identify the two factors. Are they crossed or nested? Be specific.
- List all the treatments.
- Identify the response and classify it as quantitative or categorical.
- Explain the importance of randomization in this study.

WHAT THIS QUESTION IS REALLY ASKING

Pure vocabulary, no arithmetic. He wants the words **factor**, **level**, **treatment**, **crossed**, **response**, **quantitative** used correctly about *this* study. Two factors that pair up in every combination are crossed. Randomisation matters because it stops unmeasured differences between students lining up with the device or app.

SOLUTION A1

(a) Factor 1 is **device type** with 2 levels (tablet, laptop). Factor 2 is **study app** with 3 levels (A, B, C). They are **crossed**, because every level of device type appears with every level of study app, so all 2×3 combinations exist.

CROSSED VS NESTED

Crossed: every level of one factor occurs with every level of the other, so you can estimate an interaction. **Nested**: the levels of one factor are different *within* each level of the other (e.g. different teachers within each school), so no interaction is estimable. Ask: "is App A the same App A on both devices?" Yes, so crossed.

(b) The $2 \times 3 = 6$ treatments are: tablet+App A, tablet+App B, tablet+App C, laptop+App A, laptop+App B, laptop+App C.

(c) The response is **student test score** as a percentage. It is **quantitative** (numeric and continuous).

(d) Randomly assigning participants means any uncontrolled nuisance variables, such as prior ability or motivation, are spread evenly across the six treatments rather than lining up with one of them. That removes systematic bias and is what allows a difference in scores to be attributed to the device and app rather than to a confounder.

MARKING

"Be specific" means naming the actual levels from this problem, not saying "the two factors are crossed". Every part here wants the study's own words.

A2 [5 MARKS]

Four pesticides tested on crop yield (tons). Farmers randomly assigned.

PESTICIDE j	N_j	$\bar{Y}_j$	$\sum_i Y_{ij}^2$
1	5	3.26	53.25
2	6	3.65	80.09
3	4	3.25	42.34
4	5	3.36	56.58

(a) Find $\bar{Y}_T$. (b) Find SSA. (c) Find SSE. (d) Complete the ANOVA table. (e) Test for a difference in crop yield among the four pesticides.

WHAT THIS QUESTION IS REALLY ASKING

A standard one-way ANOVA, but he has set two traps on purpose: the group sizes differ (**unbalanced**, so the grand mean must be weighted) and he gives you **sum of squared observations** instead of variances (so use the computational SS formulas, not the deviation ones). Spot both in the first 20 seconds and the question is routine.

SOLUTION A2

READ THIS FIRST

This design is **unbalanced** (5, 6, 4, 5) and gives you $\sum Y_{ij}^2$ instead of s_j^2 . Both change which formula you use. This is the exact reason he set it this way.

$$N = 5 + 6 + 4 + 5 = 20$$

$$\bar{Y}_T = \frac{5(3.26) + 6(3.65) + 4(3.25) + 5(3.36)}{20} = \frac{16.3 + 21.9 + 13.0 + 16.8}{20} = \frac{68.0}{20} = \boxed{3.4}$$

WHY Unbalanced, so you **must** weight each group mean by its n_j . The plain average of 3.26, 3.65, 3.25, 3.36 would give 3.38, which is wrong.

$$SSA = \sum n_j \bar{Y}_j^2 - N \bar{Y}_T^2$$

$$= 5(3.26)^2 + 6(3.65)^2 + 4(3.25)^2 + 5(3.36)^2 - 20(3.4)^2$$

$$= 53.138 + 79.935 + 42.25 + 56.448 - 231.2 = 231.771 - 231.2 = \boxed{0.571}$$

WHY The computational form from the sheet. It avoids subtracting the grand mean 20 times. Keep all decimals; these numbers are small and differences of large numbers, so early rounding destroys them.

$$SSE = \sum_j \sum_i Y_{ij}^2 - \sum_j n_j \bar{Y}_j^2$$

$$= (53.25 + 80.09 + 42.34 + 56.58) - 231.771 = 232.26 - 231.771 = \boxed{0.489}$$

WHY Also from the sheet. The first bracket is the raw sum of squares of every observation; the second is what the group means account for. What is left over is error. Notice $\sum n_j \bar{Y}_j^2 = 231.771$ was already computed in part (b), so reuse it.

df: $a - 1 = 3$, $N - a = 16$, $N - 1 = 19$.

$$MSA = \frac{0.571}{3} = 0.1903, \quad MSE = \frac{0.489}{16} = 0.0306, \quad F = \frac{0.1903}{0.0306} = 6.2277$$

$$SST = 0.571 + 0.489 = 1.06$$

WHY Carry the unrounded values into F: $0.190333/0.0305625 = 6.2277$.

SOURCE	DF	SS	MS	F
Treatment	3	0.571	0.1903	6.2277
Error	16	0.489	0.0306	
Total	19	1.06		

(e) Let μ_j be the mean crop yield under pesticide j.
 $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ vs H_1 : at least two differ.
 $F_{0.05, 3, 16} = 3.24$. Since $6.2277 > 3.24$, **reject** H_0 : there is evidence that mean crop yield differs among the four pesticides.

WHY Numerator df 3 (treatment), denominator 16 (error).

A3 [4 MARKS]

Three drug doses (Low, Medium, High) on blood pressure reduction. All participants follow the same diet, exercise regimen and sleep schedule. Double-blind. Randomly assigned. Balanced.

SOURCE	DF	SUM SQ	MEAN SQ	F VALUE
Treatments	?	72.5	?	?
Residuals	24	96.0	?	

- (a) Write the 4 principles of experimental design in order, and say YES or NO for whether each is used here.
 (b) Explain how a double-blind design is implemented here and its primary advantage.
 (c) Test whether blood pressure reduction differs among the three doses.

WHAT THIS QUESTION IS REALLY ASKING

Two halves. The first is Unit 1 recall: his four principles in his order. The second is an ANOVA table with blanks, where the treatment df is not given to you but is implied by the words 'three drug doses'. Read the sentence, not just the table.

SOLUTION A3

HIS 4 PRINCIPLES, AND THE ORDER MATTERS

(1) Control (2) Block (3) Randomize (4) Replicate.

The logic: if you can identify a nuisance variable and hold it constant, **control** it. If you can identify it but not hold it constant, **block** on it. If you cannot even identify it, **randomize** so it spreads evenly. And always **replicate**, because without more than one unit per treatment you have no way to estimate error.

PRINCIPLE	USED?	WHY
1. Control	YES	Diet, exercise and sleep are held constant for everyone.
2. Block	NO	No blocking variable is mentioned; participants are not grouped by anything.
3. Randomize	YES	Participants are randomly assigned to one of the three doses.
4. Replicate	YES	Balanced design with 9 participants per dose, so more than one unit per treatment.

(b) Double-blind means **neither the participant nor the person measuring the blood pressure knows which dose was given**. The advantage is that it removes bias from both sides: the participant cannot be influenced by expectation (placebo effect), and the assessor cannot unconsciously record readings in the direction they expect. So any difference is attributable to the dose itself.

(c) $a = 3$ doses so treatment $df = 2$. Residual $df = N - a = 24$, so $N = 27$ (9 per dose).

$$MSA = \frac{72.5}{2} = 36.25, \quad MSE = \frac{96.0}{24} = 4.0, \quad F_{\text{obs}} = \frac{36.25}{4.0} = 9.0625$$

WHY The dashes in the table are what you fill. Treatment df comes from the number of doses stated in the words, not from the table.

$H_0 : \mu_L = \mu_M = \mu_H$ vs at least two differ. $F_{0.05,2,24} = 3.40$. Since $9.0625 > 3.40$, **reject**: mean blood pressure reduction differs among the three doses.

WHY Define μ in words first, as always.

A4

[6 MARKS]

Teaching methods (Lecture, Group Work, Online) on test scores. Model $Y_i = \beta_0 + \beta_1 I_{1,i} + \beta_2 I_{2,i} + \epsilon_i$ where $I_1 = 1$ for Lecture, $I_2 = 1$ for Group Work, 0 otherwise.

METHOD	N_j	$\bar{Y}_j$	S_j
Lecture	25	78.5	4.1
Group Work	25	81.3	4.5
Online	25	75.8	4.2

(a) Interpret β_0 . (b) Interpret β_2 . (c) Estimate β_2 . (d) Estimate σ^2 . (e) Find SST.

WHAT THIS QUESTION IS REALLY ASKING

This is a coding question wearing a regression disguise. The whole thing turns on one observation: whichever group has **no indicator** is the reference. Once you know that, every beta is either the reference mean or a difference from it, and sigma-squared is just MSE.

SOLUTION A4

Which group is the reference? A student in **Online** has $I_1 = 0$ and $I_2 = 0$. So **Online is the reference**.

WHY Always find the reference first: it is whichever group has **no** indicator of its own. Here only Lecture and Group Work got indicators.

(a) β_0 is the **mean test score of students taught by the Online method**.

(b) β_2 is the **difference in mean test score between the Group Work method and the Online method**. Positive means Group Work scores higher.

$$\hat{\beta}_2 = \bar{Y}_{\text{GroupWork}} - \bar{Y}_{\text{Online}} = 81.3 - 75.8 = \boxed{5.5}$$

WHY Dummy coefficients are differences from the reference.

$$\begin{aligned} \hat{\sigma}^2 = MSE &= \frac{\sum (n_j - 1) s_j^2}{N - a} = \frac{24(4.1^2 + 4.5^2 + 4.2^2)}{75 - 3} \\ &= \frac{24(16.81 + 20.25 + 17.64)}{72} = \frac{24(54.70)}{72} = \frac{1312.8}{72} = \boxed{18.2333} \end{aligned}$$

WHY $\hat{\sigma}^2$ is MSE, always. They gave standard deviations, so square them.

$$\begin{aligned} \bar{Y}_T &= \frac{78.5 + 81.3 + 75.8}{3} = 78.5333 \quad (\text{balanced}) \\ SSA &= 25[(78.5 - 78.5333)^2 + (81.3 - 78.5333)^2 + (75.8 - 78.5333)^2] \\ &= 25[0.0011 + 7.6544 + 7.4711] = 25(15.1267) = 378.1667 \\ SSE &= 1312.8, \quad SST = 378.1667 + 1312.8 = \boxed{1690.9667} \end{aligned}$$

WHY SST is asked but not directly available, so build it as $SSA + SSE$. SSE was already computed in part (d) before dividing by 72, so reuse 1312.8.

A5

[3 MARKS] · THE PROOF

Three drug formulations. $H_0 : \mu_1 = \mu_2 = \mu_3$, each $Y_{ij} \sim N(\mu_j, \sigma^2)$, $N = n_1 + n_2 + n_3$. Prove that $F = \frac{MSA}{MSE}$ follows an F distribution with $a - 1$ and $N - a$ df under H_0 .

Show all steps and explicitly mention the name of the theorems or facts being used. Reasoning must appear at the correct step or the justification is not counted.

WHAT THIS QUESTION IS REALLY ASKING

A bookwork proof. He is not testing algebra, he is testing whether you can **name the facts**: divide by sigma squared, invoke Cochran for independence, then quote the definition of an F. The marks are attached to those three phrases.

SOLUTION A5

$$F = \frac{MSA}{MSE} = \frac{SSA/(a-1)}{SSE/(N-a)} = \frac{(SSA/\sigma^2)/(a-1)}{(SSE/\sigma^2)/(N-a)}$$

WHY **This division by σ^2 is worth a mark on its own.** His Test 1 key writes in the margin: "here the student must divide by σ^2 . If not, minus 1 point." It changes nothing numerically but it converts each sum of squares into a chi-square.

Under H_0 : $\frac{SSA}{\sigma^2} \sim \chi^2(a-1)$ and $\frac{SSE}{\sigma^2} \sim \chi^2(N-a)$, and by **Cochran's theorem** they are **independent**.

WHY Name both facts. Cochran's theorem is precisely what gives independence of the two quadratic forms, and independence is required by the definition of F.

$$\stackrel{d}{=} \frac{\chi^2(a-1)/(a-1)}{\chi^2(N-a)/(N-a)} \sim F_{a-1, N-a}$$

WHY By **definition**, an F random variable is the ratio of two independent chi-squares each divided by its own degrees of freedom. Say the word "by definition".

MARKS ARE FOR THE WORDS

Three named justifications: **divide by sigma squared**, **by Cochran's theorem**, **independent**, and **by definition of the F distribution**. The algebra alone is not worth full marks.

PAPER B · STA305 TERM TEST 2, MARCH 2025 · 20 MARKS

Full past paper

B1

[7 MARKS]

Five diets (A to E) on fish growth. 20 fish randomly and evenly assigned. Weight in grams.

means: A 50.4 B 48.5 C 49.2 D 52.4 E 46.4

Analysis of Variance Table

Response: Weight

	Df	Sum Sq	Mean Sq	F value
Type	-	79.2	-	-
Residuals	-	66.9	-	-

(a) Test whether mean fish weight differs among the five diets.

(b) List the three required ANOVA assumptions. Say YES or NO for each and give the p-value where applicable, from this output. Write N/A if no p-value applies.

```
shapiro_test(residuals(regmodel.dummy))
  statistic p.value
    0.951    0.390

bartlett.test(Weight ~ Diet)
Bartlett's K-squared = 0.52679, df = 4, p-value = 0.9708
```

(c) Would you recommend the Kruskal-Wallis test instead of ANOVA? YES or NO, and justify.

(d) Pairwise comparisons, unadjusted:

	A	B	C	D
B	0.2230	-	-	-
C	0.4367	0.6435	-	-
D	0.2110	0.0210	0.0525	-
E	0.0163	0.1724	0.0761	0.0011

All pairwise tests at the 5% family-wise level using Sidak-Bonferroni. (i) Derive the formula for α_{SB} . (ii) Is there evidence that diets D and E differ? YES or NO, with the p-value and all steps.

WHAT THIS QUESTION IS REALLY ASKING

Four separate skills bolted onto one dataset: fill an R table, match each diagnostic test to the assumption it checks, decide whether to abandon ANOVA, and apply a family-wise correction. The correction part hinges on counting **all** pairwise comparisons, not the single one he asks about.

SOLUTION B1

(a) 20 fish evenly over 5 diets, so $n_j = 4$, $N = 20$, $a = 5$. df: $a - 1 = 4$, $N - a = 15$, total 19.

$$MSA = \frac{79.2}{4} = 19.8, \quad MSE = \frac{66.9}{15} = 4.46, \quad F_{\text{obs}} = \frac{19.8}{4.46} = 4.4395$$

WHY The Df column is blank, so you recover it from the words: five diets and twenty fish.

$H_0: \mu_A = \mu_B = \mu_C = \mu_D = \mu_E$ vs at least two differ, where μ_j is the mean weight under diet j.

$F_{0.05, 4, 15} = 3.06$. Since $4.4395 > 3.06$, **reject**: mean fish weight differs among the five diets.

WHY Numerator 4, denominator 15.

MATCHING EACH TEST TO ITS ASSUMPTION

Shapiro-Wilk tests **normality** of the residuals. **Bartlett** (or Levene) tests **equal variance**. **Independence** has no test at all: you argue it from how the data were collected. Large p-value means the assumption is *not* contradicted, so YES it is satisfied.

ASSUMPTION	SATISFIED?	P-VALUE	REASON
1. Independence	YES	N/A	No test exists. The fish were randomly and independently assigned, so it is justified by design.
2. Normality of errors	YES	0.390	Shapiro-Wilk, $0.390 > 0.05$, so no evidence against normality.
3. Equal variance	YES	0.9708	Bartlett, $0.9708 > 0.05$, so no evidence against equal variances.

(c) NO. All three ANOVA assumptions are satisfied, so there is no reason to abandon the parametric test. Kruskal-Wallis is only preferred when normality or equal variance fail. Parametric methods are **always more powerful when their assumptions hold**, so switching would only lose power.

(d)(i) For k independent tests each at level α_{SB} , the chance of no false positive anywhere is $(1 - \alpha_{SB})^k$, so

$$\alpha_{FW} = 1 - (1 - \alpha_{SB})^k \implies (1 - \alpha_{SB})^k = 1 - \alpha_{FW}$$

$$\implies \alpha_{SB} = 1 - (1 - \alpha_{FW})^{1/k}$$

WHY He asked you to **derive** it, not quote it. Start from the family-wise rate, invert. This is the only derivation in the post-hoc material and it is two lines.

(ii) All pairwise among 5 diets: $k = \binom{5}{2} = 10$.

$$\alpha_{SB} = 1 - (1 - 0.05)^{1/10} = 1 - (0.95)^{0.1} = 1 - 0.9949 = 0.0051$$

D vs E has p-value 0.0011. Since $0.0011 < 0.0051$, **reject**. Answer: **YES**, diets D and E differ significantly.

WHY k counts **all** ten pairwise comparisons, not the single one you were asked about. Compare with Bonferroni: $0.05/10 = 0.005$, almost identical but slightly stricter, which is why Sidak has marginally more power.

B2

[4 MARKS]

Continuation of B1. Three coding models for the fish data:

$$\text{Dummy: } E[Y] = \beta_0 + \beta_1 I_B + \beta_2 I_C + \beta_3 I_D + \beta_4 I_E$$

$$\text{Effect: } E[Y] = \alpha_0 + \alpha_1 X_A + \alpha_2 X_B + \alpha_3 X_C + \alpha_4 X_D$$

$$\text{Cell means: } E[Y] = \gamma_1 I_A + \gamma_2 I_B + \gamma_3 I_C + \gamma_4 I_D + \gamma_5 I_E$$

where each X is 1 for its own diet, -1 for diet E, 0 otherwise.

(a) Estimate β_0 . (b) Estimate α_0 . (c) Express $H_0 : \beta_4 = 0$ in terms of the diet means. (d) Write H_0 of equal means for the **cell means** model.

WHAT THIS QUESTION IS REALLY ASKING

The same three coding schemes again, so recognise the pattern rather than re-deriving. The only real thinking is spotting that diet A is the reference (it has no indicator) and that effect coding centres on the grand mean.

SOLUTION B2

(a) Indicators exist for B, C, D, E, so **A is the reference**.

$$\hat{\beta}_0 = \bar{Y}_A = \boxed{50.4}$$

WHY In dummy coding the intercept is the reference group's mean. Find the reference by seeing which diet has no indicator.

(b)

$$\hat{\alpha}_0 = \bar{Y}_T = \frac{50.4 + 48.5 + 49.2 + 52.4 + 46.4}{5} = \frac{246.9}{5} = \boxed{49.38}$$

WHY In effect coding the intercept is the **grand mean**, not any group's mean. Balanced ($n_j = 4$ each), so the plain average of the five means is the grand mean.

(c) β_4 multiplies I_E , so $\beta_4 = \mu_E - \mu_A$. Hence

$$H_0 : \mu_E - \mu_A = 0, \quad \text{that is} \quad \boxed{H_0 : \mu_E = \mu_A}$$

WHY Translating betas back into means. Dummy coefficients are always differences from the reference.

(d) $\boxed{H_0 : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5}$

THE CLASSIC MIX-UP

Cell means gammas **are** the means, so you set them **equal to each other**. Dummy and effect coefficients are **differences**, so you set them **equal to zero**. Getting this backwards is the most common lost mark in this whole topic.

B3

[5 MARKS]

Four training programs, improvement scores. $n_j = 5$ each.

PROGRAM	N_j	$\bar{Y}_j$	S_j
1	5	12.5	1.6
2	5	11	2.3
3	5	9.5	1.3
4	5	9.2	2.8

$$\psi_1 = \mu_1 - \mu_2, \psi_2 = \mu_3 - \mu_4, \psi_3 = \mu_1 + \mu_2 - \mu_3 - \mu_4.$$

(a) Are they mutually orthogonal? (b) Test $H_0 : \psi_1 = 0$ at 5%. (c) Using Bonferroni with all three contrasts, build an **85%** CI for ψ_3 .

WHAT THIS QUESTION IS REALLY ASKING

Three contrasts, so 'mutually orthogonal' means checking **three pairs**, not one. Then one contrast is tested alone (no correction) and another gets a corrected interval (correction needed). He is checking that you know when a correction applies and when it does not.

SOLUTION B3

(a) Coefficients: $\mathbf{c}_1 = (1, -1, 0, 0)$, $\mathbf{c}_2 = (0, 0, 1, -1)$, $\mathbf{c}_3 = (1, 1, -1, -1)$. Each sums to zero. Balanced, so use plain dot products, and check **all three pairs**:

$$\mathbf{c}_1 \cdot \mathbf{c}_2 = 0 + 0 + 0 + 0 = 0$$

$$\mathbf{c}_1 \cdot \mathbf{c}_3 = (1)(1) + (-1)(1) + 0 + 0 = 0$$

$$\mathbf{c}_2 \cdot \mathbf{c}_3 = 0 + 0 + (1)(-1) + (-1)(-1) = -1 + 1 = 0$$

All zero, so **YES**, mutually orthogonal.

WHY Orthogonality is a property of a **pair**, so with 3 contrasts you must check $\binom{3}{2} = 3$ pairs. Missing one loses the mark. Note also $a - 1 = 3$, the maximum possible size of a mutually orthogonal set, so these three split SSA completely.

(b)

$$SSE = \sum (n_j - 1)s_j^2 = 4(1.6^2 + 2.3^2 + 1.3^2 + 2.8^2) = 4(2.56 + 5.29 + 1.69 + 7.84) = 4(17.38) = 69.52$$

$$N = 20, a = 4, \quad MSE = \frac{69.52}{16} = 4.345$$

WHY Standard deviations again, so square first. Error df = $N - a = 16$.

$$\hat{\psi}_1 = 12.5 - 11 = 1.5, \quad \sum \frac{c_j^2}{n_j} = \frac{1+1}{5} = 0.4$$

$$F_{\text{obs}} = \frac{\hat{\psi}_1^2}{MSE \sum c_j^2 / n_j} = \frac{2.25}{4.345 \times 0.4} = \frac{2.25}{1.738} = \boxed{1.2946}$$

$F_{0.05, 1, 16} = 4.49$. Since $1.2946 < 4.49$, **FTR**: no evidence that programs 1 and 2 differ.

WHY A contrast always has numerator df 1. The hint says no correction is needed because it is a single test.

(c) 85% confidence means $\alpha_{FW} = 0.15$. Three contrasts, so $k = 3$:

$$\alpha_B = \frac{0.15}{3} = 0.05 \Rightarrow t_{\alpha_B/2, 16} = t_{0.025, 16} = 2.120$$

WHY Bonferroni divides alpha by k. Then because the interval is two-sided you halve it **again** for the t lookup: 0.05 becomes 0.025.

$$\hat{\psi}_3 = 12.5 + 11 - 9.5 - 9.2 = 4.8, \quad \sum \frac{c_j^2}{n_j} = \frac{1+1+1+1}{5} = 0.8$$

$$s_{\hat{\psi}_3} = \sqrt{4.345 \times 0.8} = \sqrt{3.476} = 1.8644$$

$$4.8 \pm 2.120(1.8644) = 4.8 \pm 3.9525 = \boxed{(0.8475, 8.7525)}$$

WHY Zero is **outside** the interval, so ψ_3 is significant at the family level: programs 1 and 2 together differ from programs 3 and 4 together.

B4

[4 MARKS] · THE PROOF

One-way model $y_{ij} = \mu_j + \varepsilon_{ij}$, $\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$, **equal** sample sizes n . $\bar{Y}_j = \frac{1}{n} \sum_i Y_{ij}$, $\bar{Y} = \frac{1}{na} \sum_j \sum_i Y_{ij}$.

You may use without proof: $\bar{Y}_j \sim N(\mu_j, \sigma^2/n)$ and $\bar{Y} \sim N(\mu, \sigma^2/(na))$.

(a) Derive $\text{Cov}(\bar{Y}_j, \bar{Y})$. (b) Derive $\text{Corr}(\bar{Y}_j, \bar{Y})$. (c) Derive $V(\bar{Y}_j - \bar{Y})$.

WHAT THIS QUESTION IS REALLY ASKING

A covariance proof. The key insight is that a group mean and the grand mean **share data**, so they are not independent and you cannot just add variances. Write the grand mean as an average of the group means and the whole thing collapses.

SOLUTION B4

THE KEY REALISATION

$\bar{Y}_j$ and $\bar{Y}$ are **not independent**, because $\bar{Y}$ contains every observation that $\bar{Y}_j$ contains. Whenever one mean is nested inside another, you cannot just add variances. Write the big one as an average of the small ones and the covariance collapses.

(a) Write the grand mean as the average of the group means (valid because the design is balanced):

$$\bar{Y} = \frac{1}{a} \sum_{k=1}^a \bar{Y}_k$$

$$\text{Cov}(\bar{Y}_j, \bar{Y}) = \text{Cov}\left(\bar{Y}_j, \frac{1}{a} \sum_{k=1}^a \bar{Y}_k\right) = \frac{1}{a} \sum_{k=1}^a \text{Cov}(\bar{Y}_j, \bar{Y}_k)$$

WHY Covariance is linear in each argument; that is the bilinearity fact on the formula sheet.

By independence of different treatment groups, $\text{Cov}(\bar{Y}_j, \bar{Y}_k) = 0$ whenever $k \neq j$, so only the $k = j$ term survives:

$$= \frac{1}{a} \text{Cov}(\bar{Y}_j, \bar{Y}_j) = \frac{1}{a} V(\bar{Y}_j) = \frac{1}{a} \cdot \frac{\sigma^2}{n} = \boxed{\frac{\sigma^2}{na}}$$

WHY Different groups use disjoint sets of observations, hence independent, hence zero covariance. Say "by independence" here: that is where the mark is. And $\text{Cov}(X, X) = V(X)$.

(b)

$$\begin{aligned} \text{Corr}(\bar{Y}_j, \bar{Y}) &= \frac{\text{Cov}(\bar{Y}_j, \bar{Y})}{\sqrt{V(\bar{Y}_j)} \sqrt{V(\bar{Y})}} = \frac{\sigma^2/(na)}{\sqrt{\sigma^2/n} \sqrt{\sigma^2/(na)}} \\ &= \frac{\sigma^2/(na)}{\frac{\sigma}{\sqrt{n}} \cdot \frac{\sigma}{\sqrt{na}}} = \frac{\sigma^2/(na)}{\frac{\sigma^2}{n\sqrt{a}}} = \frac{n\sqrt{a}}{na} = \boxed{\frac{1}{\sqrt{a}}} \end{aligned}$$

WHY Everything cancels except a . Sanity check: with $a = 1$ the two means are the same thing and the correlation is 1. With many groups each one contributes less to the grand mean, so the correlation falls like $1/\sqrt{a}$.

(c) Using $V(X - Y) = V(X) + V(Y) - 2\text{Cov}(X, Y)$:

$$\begin{aligned} V(\bar{Y}_j - \bar{Y}) &= \frac{\sigma^2}{n} + \frac{\sigma^2}{na} - 2 \cdot \frac{\sigma^2}{na} = \frac{\sigma^2}{n} - \frac{\sigma^2}{na} \\ &= \frac{\sigma^2}{n} \left(1 - \frac{1}{a}\right) = \boxed{\frac{(a-1)\sigma^2}{na}} \end{aligned}$$

WHY The covariance term is what makes this smaller than the naive $\sigma^2/n + \sigma^2/(na)$. Notice the covariance happened to equal $V(\bar{Y})$, which always happens when one mean sits inside the other.

PAPER C · STA305 FINAL EXAM, APRIL 2023 · 40 MARKS, 3 HOURS

Full past final

Same instructor, same format, formula sheet supplied. The closest thing to your paper.

C1

[9 MARKS]

Three treatments. $n_j = 5$ each; means 62, 66, 52; **variances** 27.5, 26.5, 31.

(a) SSA. (b) SSE. (c) Complete the ANOVA table. (d) Test at 5%. (e) R^2 and interpret it, **without** the words low, medium or strong.

(f) Dummy model $E[Y] = \beta_0 + \beta_1 I_2 + \beta_2 I_3$ (treatment 1 is reference): (i) estimate β_1 ; (ii) write H_0 of equal means in the betas.

(g) Effect model $E[Y] = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2$ (treatment 3 coded -1): (i) estimate α_0 ; (ii) estimate α_2 .

WHAT THIS QUESTION IS REALLY ASKING

The bread-and-butter question, worth 9 marks. Half of it is the ANOVA table; the other half is reading coefficients off two coding schemes. Note he bans the words low/medium/strong for the R-squared, so he wants the **meaning**, not a label.

SOLUTION C1

Balanced, $N = 15$, $\bar{Y}_T = (62 + 66 + 52)/3 = 60$.

$$SSA = 5(62 - 60)^2 + 5(66 - 60)^2 + 5(52 - 60)^2 = 20 + 180 + 320 = \boxed{520}$$

$$SSE = 4(27.5) + 4(26.5) + 4(31) = 110 + 106 + 124 = \boxed{340}$$

WHY The multiplier **5** in SSA is n_j ; the multiplier **4** in SSE is $n_j - 1$. Here they gave **variances**, so no squaring.

SOURCE	DF	SS	MS	F
Treatment	2	520	260	9.1765
Error	12	340	28.3333	
Total	14	860		

(d) $H_0 : \mu_1 = \mu_2 = \mu_3$, $F_{0.05,2,12} = 3.89$, and $9.1765 > 3.89$, so **reject**: at least one treatment mean differs.

WHY Numerator 2, denominator 12.

(e)

$$R^2 = \frac{520}{860} = \boxed{0.6047}$$

60.47% of the total variability in the response is explained by which treatment the unit received.

WHY He banned the adjectives, so give the proportion and its meaning and stop. This is the same $R^2 = SSA/SST$ used as an effect size in Unit 7.

(f) $\hat{\beta}_1 = \bar{Y}_2 - \bar{Y}_1 = 66 - 62 = \boxed{4}$; $H_0 : \beta_1 = \beta_2 = 0$.

(g) $\hat{\alpha}_0 = \bar{Y}_T = \boxed{60}$; $\hat{\alpha}_2 = \bar{Y}_2 - \bar{Y}_T = 66 - 60 = \boxed{6}$.

WHY Dummy: difference from reference. Effect: intercept is the grand mean, slopes are deviations from it. **Free check**: the omitted group must return as $\mu_3 = \alpha_0 - (\alpha_1 + \alpha_2) = 60 - (2 + 6) = 52$, which matches the data.

C2

[5 MARKS]

Continues C1. $\psi_1 = \mu_1 + \mu_2 - 2\mu_3$, $\psi_2 = \mu_1 - 2\mu_2 + \mu_3$.

(a) Are they orthogonal? (b)(i) ψ_1 . (ii) SS_{ψ_1} . (c) Bonferroni **90%** CI for ψ_1 .

WHAT THIS QUESTION IS REALLY ASKING

Contrasts on data you already analysed. Orthogonality is one dot product. The confidence interval is the only part with a twist: 90% with two contrasts means you halve alpha for Bonferroni, then halve again for a two-sided t.

SOLUTION C2

(a)

$$(1)(1) + (1)(-2) + (-2)(1) = 1 - 2 - 2 = -3 \neq 0 \Rightarrow \text{NO}$$

WHY Balanced, so the plain dot product. Both do sum to zero individually, so they *are* contrasts, just not orthogonal ones.

(b)

$$\hat{\psi}_1 = 62 + 66 - 2(52) = \boxed{24}, \quad \sum \frac{c_j^2}{n_j} = \frac{1+1+4}{5} = 1.2$$

$$SS_{\psi_1} = \frac{24^2}{1.2} = \frac{576}{1.2} = \boxed{480}$$

WHY Note $c_3 = -2$ so $c_3^2 = 4$.

(c) $\alpha_{FW} = 0.10$, $k = 2$, so $\alpha_B = 0.05$ and $t_{0.025,12} = 2.179$.

$$s_{\hat{\psi}_1} = \sqrt{28.3333 \times 1.2} = \sqrt{34} = 5.8310$$

$$24 \pm 2.179(5.8310) = 24 \pm 12.7057 = \boxed{(11.2943, 36.7057)}$$

WHY Zero is outside, so ψ_1 is significant at the family level.

C3

[7 MARKS]

Two-factor experiment, A has 3 levels, B has 3 levels, $n = 2$ per cell.

	B1	B2	B3
a1	500, 580	540, 460	480, 400
a2	460, 540	560, 620	420, 480
a3	560, 600	600, 580	480, 410

Given: $MS_A = 3050$, $SS_B = 45300$, $SS_{AB} = 11200$, $SSE = 19850$.

(a) Complete the ANOVA table. (b) Find $(\alpha\beta)_{32}$. (c) Test the interaction. (d) Explain interaction in plain English **without** using the word. (e) Test both main effects.

WHAT THIS QUESTION IS REALLY ASKING

A two-way table with mixed givens: some sums of squares, one **mean square**. Convert before you start. Then the order is fixed: interaction first, and only if it is not significant may you talk about main effects.

SOLUTION C3

TWO-WAY VOCABULARY, ONCE

$\bar{Y}_{jk}$ is a **cell mean** (one A level and one B level). $\bar{Y}_{Aj}$ is a **row marginal mean**, averaging over all of B. $\bar{Y}_{Bk}$ is a **column marginal mean**. The interaction estimate is $(\alpha\beta)_{jk} = \bar{Y}_{jk} - \bar{Y}_{Aj} - \bar{Y}_{Bk} + \bar{Y}_T$, which reads as **what the cell actually did minus what pure addition predicted**.

$N = 18$. df: A 2, B 2, AB $(3 - 1)(3 - 1) = 4$, Error $N - ab = 18 - 9 = 9$, Total 17. From $MS_A = 3050$ with 2 df, $SS_A = 6100$.

WHY Error df is $N - ab$, not $N - a - b$. They gave one MS and two SS deliberately, so work backwards with $SS = MS \times df$.

SOURCE	DF	SS	MS	F
Factor A	2	6100	3050	1.3829
Factor B	2	45300	22650	10.2695
Interaction	4	11200	2800	1.2695
Error	9	19850	2205.5556	
Total	17	82450		

(b) Cell means: a3b1 580, a3b2 590, a3b3 445; a1b2 500, a2b2 590.

$$\bar{Y}_{32} = \frac{600 + 580}{2} = 590, \quad \bar{Y}_{A3} = \frac{580 + 590 + 445}{3} = 538.3333$$

$$\bar{Y}_{B2} = \frac{500 + 590 + 590}{3} = 560, \quad \bar{Y}_T = 515$$

$$(\alpha\beta)_{32} = 590 - 538.3333 - 560 + 515 = \boxed{6.6667}$$

WHY Subscript 32 means level 3 of A and level 2 of B. Marginals average the cell means in that row or column.

(c) $H_0 : (\alpha\beta)_{jk} = 0$ for all j,k. $F = 1.2695$ vs $F_{0.05,4,9} = 3.63 \rightarrow \mathbf{FTR}$.

WHY Interaction is **always tested first**. Only because it is not significant may you go on and read the main effects.

(d) The effect of Factor A on the response is **the same at every level of Factor B**. The difference between two levels of A does not depend on which level of B you are in, so the two factors act separately and their effects simply add.

(e) Both against $F_{0.05,2,9} = 4.26$.

A: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$, $1.3829 < 4.26 \rightarrow \mathbf{FTR}$.

B: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$, $10.2695 > 4.26 \rightarrow \mathbf{reject}$, Factor B affects the response.

WHY Same MSE denominator for all three tests.

C4

[7 MARKS]

Balanced incomplete block design. Rows are treatments, columns are blocks; blank means not used.

TREATMENT	BLK 1	BLK 2	BLK 3	BLK 4
1	14	17		13
2		14	14	13
3	12	12	13	
4	13		11	12

(a) Find a, b, k, r, N, λ . (b) Complete the T_i, B_i, Q_i table. (c) Find SSA (adjusted). (d) Find SSB . (e) Test the treatment effect at 5%, given $MSE = 1.8577$.

WHAT THIS QUESTION IS REALLY ASKING

The most mechanical question on the paper once you know the recipe, and the most confusing if you do not. Everything flows from reading a, b, k and r off the table by **counting blanks**. The Q values summing to zero is your safety net.

SOLUTION C4

BIBD VOCABULARY, ONCE

k = block size, how many treatments fit in one block. **Incomplete** means $k.r = \text{how many blocks each treatment appears in}$. $\lambda = \text{how many blocks contain any given pair of treatments}$. Why ad just? In a complete design every treatment meet every block so block effects cancel out. Here they do not, so at treatment

(a) Four treatment rows, four block columns. Each column holds 3 numbers so $k = 3$; each row holds 3 numbers so $r = 3$.

$$a = 4, b = 4, k = 3, r = 3, N = ar = 12, \lambda = \frac{r(k-1)}{a-1} = \frac{3(2)}{3} = 2$$

WHY Count the blanks, do not guess. Check $N = bk = 4(3) = 12$ agrees with $ar = 12$.

Treatment totals (across each row): $T_1 = 44, T_2 = 41, T_3 = 37, T_4 = 36$.

Block totals (down each column): 39, 43, 38, 38. Grand total 158.

WHY Both must add to 158. They do, so no arithmetic slip yet.

TREATMENT	1	2	3	4
T_i	44	41	37	36
appears in blocks	1, 2, 4	2, 3, 4	1, 2, 3	1, 3, 4
B_i	$39+43+38 = 120$	$43+38+38 = 119$	$39+43+38 = 120$	$39+38+38 = 115$
$Q_i = T_i - B_i/3$	4	1.3333	-3	-2.3333

Check: $4 + 1.3333 - 3 - 2.3333 = 0 \checkmark$

WHY **The Q_i always sum to zero.** This single check catches almost every error in this question, usually a mis-assigned block in B_i .

(c)

$$\sum Q_i^2 = 16 + 1.7778 + 9 + 5.4444 = 32.2222$$

$$SSA_{\text{adj}} = \frac{k \sum Q_i^2}{\lambda a} = \frac{3(32.2222)}{2 \times 4} = \frac{96.6667}{8} = \boxed{12.0833}$$

WHY Straight from the sheet, with $k = 3$, $\lambda = 2$, $a = 4$.

(d) Block means: $39/3 = 13$, $43/3 = 14.3333$, $38/3 = 12.6667$, $38/3 = 12.6667$; $\bar{y}_T = 158/12 = 13.1667$.

$$SSB = \sum k \bar{y}_{Bj}^2 - N \bar{y}_T^2 = 3(13^2 + 14.3333^2 + 12.6667^2 + 12.6667^2) - 12(13.1667)^2$$

$$= 3(695.3333) - 12(173.3611) = 2086 - 2080.3333 = \boxed{5.6667}$$

WHY Blocks are **not** adjusted; only treatments are. The multiplier $k = 3$ is because each block mean summarises 3 observations.

(e) H_0 : there is no treatment effect ($\alpha_1 = \dots = \alpha_4 = 0$) vs at least one non-zero.

$$MSA = \frac{12.0833}{3} = 4.0278, \quad F_{\text{obs}} = \frac{4.0278}{1.8577} = \boxed{2.1682}$$

Error df = $N - a - b + 1 = 12 - 4 - 4 + 1 = 5$, so $F_{0.05,3,5} = 5.41$. Since $2.1682 < 5.41$, **FTR: no evidence the treatments differ.**

WHY BIBD error df has the same shape as RCB. He supplied MSE so you did not need SSE.

C5

[8 MARKS] · THE BIGGEST QUESTION ON THE PAPER

Three diets A, B, C on weight loss Y , controlling for age x . Diet A is the reference. β_1 is the coefficient of x .

(a) Write the parallel planes model with dummy coding, defining every term.

(b) In terms of the betas, write H_0 for: (i) controlling for age, mean weight loss is the same across all diets; (ii) controlling for age, A and B together equal C; (iii) weight loss is not affected by age.

(c) Write the model allowing the age effect to vary by diet.

(d) In terms of the betas, write H_0 for: (i) interaction between diet and covariate; (ii) equal slopes for all diets; (iii) controlling for age, diet has no significant effect.

WHAT THIS QUESTION IS REALLY ASKING

No arithmetic at all, 8 marks. It is entirely about writing hypotheses in terms of betas. Build the table of group rows first and every answer is read off it by subtraction. Never try to recall these.

SOLUTION C5

ANCOVA, AND THE ONLY TECHNIQUE YOU NEED

A **covariate** is a numeric variable you measured but did not assign. Controlling for it does the same job as blocking: it takes explainable variation out of the error term. **Parallel planes** = every group shares one slope; the group terms only shift the intercept. **Never recall an ANCOVA hypothesis. Build the table of group rows and read it off.**

(a) Let $I_B = 1$ if the subject is on diet B and 0 otherwise; $I_C = 1$ if on diet C and 0 otherwise. Diet A is the reference.

$$E[Y|X = x] = \beta_0 + \beta_1 x + \beta_2 I_B + \beta_3 I_C$$

WHY Two indicators for three groups. **Defining them in words is worth a mark.**

DIET	I_B	I_C	$E[Y X = x]$
A (ref)	0	0	$\beta_0 + \beta_1 x$
B	1	0	$(\beta_0 + \beta_2) + \beta_1 x$
C	0	1	$(\beta_0 + \beta_3) + \beta_1 x$

(b)(i) $H_0 : \beta_2 = \beta_3 = 0$

WHY Setting both to zero makes all three rows identical, which is exactly "same mean at every age".

(ii) "A and B together equal C" means $\frac{\mu_A + \mu_B}{2} = \mu_C$:

$$\frac{\beta_0 + (\beta_0 + \beta_2)}{2} = \beta_0 + \beta_3 \Rightarrow \beta_0 + \frac{\beta_2}{2} = \beta_0 + \beta_3 \Rightarrow \boxed{H_0 : \beta_2 = 2\beta_3}$$

WHY Average the two rows, set equal to the third. β_0 and $\beta_1 x$ appear on both sides and cancel. Pure algebra off the table, nothing memorised.

(iii) $\boxed{H_0 : \beta_1 = 0}$

WHY β_1 is the only place x appears.

(c)

$$E[Y|X = x] = \beta_0 + \beta_1 x + \beta_2 I_B + \beta_3 I_C + \beta_4 x I_B + \beta_5 x I_C$$

WHY One extra term per non-reference group: x times that group's indicator.

DIET	$E[Y X = X]$	SLOPE
A (ref)	$\beta_0 + \beta_1 x$	β_1
B	$(\beta_0 + \beta_2) + (\beta_1 + \beta_4)x$	$\beta_1 + \beta_4$
C	$(\beta_0 + \beta_3) + (\beta_1 + \beta_5)x$	$\beta_1 + \beta_5$

(d)(i) and (ii) are the same hypothesis: $\boxed{H_0 : \beta_4 = \beta_5 = 0}$

WHY Read the slope column: the three slopes are equal exactly when $\beta_4 = \beta_5 = 0$. **Equal slopes IS no interaction**, so one answer serves both parts.

(iii) $\boxed{H_0 : \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0}$

WHY "Diet has no effect at all" means every group row collapses onto the reference row, so both the intercept shifts and the slope shifts must vanish. All four.

THE TWO THAT GET SWAPPED

"A and B are the same **as each other**" is $\beta_2 = \beta_3$. "A and B **together** equal the third group" is an average, giving $\beta_2 = 2\beta_3$ here. Read the sentence twice, then derive from the table.

C6

[4 MARKS] · THE PROOF

Two-way model with **one observation per cell**: $y_{ij} = \mu_T + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij}$, $\varepsilon_{ij} \sim N(0, \sigma^2)$ independent, with $\sum_i \alpha_i = \sum_j \beta_j = \sum_i (\alpha\beta)_{ij} = \sum_j (\alpha\beta)_{ij} = 0$.

$$\bar{y}_i = \frac{1}{b} \sum_j y_{ij}, \bar{y}_T = \frac{1}{ab} \sum_i \sum_j y_{ij}.$$

(a) Derive the distribution of $\bar{y}_i$. (b) Derive $V(\bar{y}_T)$. (c) Derive $V(\bar{y}_i - \bar{y}_T)$. *Hint given on the paper: $V(aX + bY) = a^2V(X) + b^2V(Y) + 2ab \text{Cov}(X, Y)$.*

WHAT THIS QUESTION IS REALLY ASKING

Same shape as B4 but in the two-way model. The constraints that all effects sum to zero are what kill most of the terms; the difficulty is again the covariance between a mean and the grand mean.

SOLUTION C6

(a)

$$\bar{y}_i = \frac{1}{b} \sum_{j=1}^b (\mu_T + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ij})$$

μ_T and α_i do not depend on j , so each averages to itself. **By the stated constraints** $\sum_j \beta_j = 0$ and $\sum_j (\alpha\beta)_{ij} = 0$, so both vanish.

$$E[\bar{y}_i] = \mu_T + \alpha_i$$

WHY Averaging over j is exactly what triggers the j constraints. Name the constraint explicitly.

$$V(\bar{y}_i) \stackrel{\text{by independence}}{=} \frac{1}{b^2} \sum_{j=1}^b V(\varepsilon_{ij}) = \frac{b\sigma^2}{b^2} = \frac{\sigma^2}{b}$$

A linear combination of independent normals is normal, so $\boxed{\bar{y}_i \sim N(\mu_T + \alpha_i, \sigma^2/b)}$.

WHY The $1/b$ comes out **squared** because variance is a squared quantity. Only the errors carry variance; the fixed effects are constants.

(b)

$$V(\bar{y}_T) \stackrel{\text{by independence}}{=} \frac{1}{(ab)^2} \sum_i \sum_j V(\varepsilon_{ij}) = \frac{ab\sigma^2}{(ab)^2} = \boxed{\frac{\sigma^2}{ab}}$$

WHY Same move with ab observations. All fixed effects sum to zero over both indices, so nothing survives except the errors.

(c) These are **not** independent: $\bar{y}_T$ contains every observation in $\bar{y}_i$. Write $\bar{y}_T = \frac{1}{a} \sum_k \bar{y}_k$. Distinct $\bar{y}_k$ use disjoint errors, so **by independence** only the $k = i$ term survives:

$$\text{Cov}(\bar{y}_i, \bar{y}_T) = \frac{1}{a} V(\bar{y}_i) = \frac{1}{a} \cdot \frac{\sigma^2}{b} = \frac{\sigma^2}{ab}$$

WHY This is the whole difficulty, and it is why the hint on the paper is the $V(aX + bY)$ formula. **When one mean is nested inside another, their covariance equals the variance of the bigger one.**

$$V(\bar{y}_i - \bar{y}_T) = \frac{\sigma^2}{b} + \frac{\sigma^2}{ab} - 2 \cdot \frac{\sigma^2}{ab} = \frac{\sigma^2}{b} - \frac{\sigma^2}{ab} = \frac{(a-1)\sigma^2}{ab}$$

WHY Apply the hint with coefficients 1 and -1 , then factor. Compare with B4(c), which is the identical structure in the one-way case.

PAPER D • YOUR OWN TEST 1 AND TEST 2, SUMMER 2026

The style he is using this term

These are from the two tests you actually wrote. The final is "aligned with Test 1 and Test 2 in style and expectations", in his words.

D1

[3 MARKS] • TEST 2 Q2

28 adults, 7 per routine, four evening routines. Means A 6.4, B 7.1, C 6.8, D 7.7.

	Df	Sum Sq	Mean Sq	F value
Routine	3	6.30	?	?
Residuals	?	11.20	?	

(a) Test for a difference in mean sleep duration. (b) Find R^2 and interpret with Cohen's guidelines.

WHAT THIS QUESTION IS REALLY ASKING

R output with blanks. Recover a from the factor df, N from the words, and remember R never prints a Total row, so you build SST yourself.

SOLUTION D1

$a = 3 + 1 = 4$, $N = 28$, residual df = 24.

$$MSA = \frac{6.30}{3} = 2.10, \quad MSE = \frac{11.20}{24} = 0.4667, \quad F = \frac{2.10}{0.4667} = 4.5$$

$F_{0.05, 3, 24} = 3.01$, and $4.5 > 3.01$, so **reject**: evening routine affects mean sleep duration.

WHY Recover a from the factor Df, then residual df = $N - a$.

$$R^2 = \frac{6.30}{6.30 + 11.20} = \frac{6.30}{17.50} = 0.36 \Rightarrow \text{large}$$

WHY R does not print a Total row, so you build $SST = SSA + SSE$ yourself. Cohen's large threshold is $R^2 \approx 0.15$. Here he **did** want the label, unlike C1(e) where he banned it. Read the wording.

D2

[5 MARKS] • TEST 2 Q3

Continues D1. Dummy uses A as reference; effect coding uses D as the -1 group, with X_1 flagging A, X_2 flagging B, X_3 flagging C.

(a) Estimate β_2 . (b) Estimate α_3 . (c) H_0 of equal means in the cell means model. (d) Same in the dummy model. (e) If $H_0 : \beta_3 = 0$, rewrite in terms of population means.

WHAT THIS QUESTION IS REALLY ASKING

Coding again. The single most common error here is assuming the subscript on beta matches the group number. Map the indicators explicitly before computing anything.

SOLUTION D2

Map the subscripts first: $\beta_1 \rightarrow I_B$, $\beta_2 \rightarrow I_C$, $\beta_3 \rightarrow I_D$.

$$\hat{\beta}_2 = \bar{Y}_C - \bar{Y}_A = 6.8 - 6.4 = \boxed{0.4}$$

WHY **The subscript is not the group number.** With A as reference the first indicator is B, so β_2 is about C.

$$\bar{Y}_T = \frac{6.4 + 7.1 + 6.8 + 7.7}{4} = 7.0, \quad \hat{\alpha}_3 = \bar{Y}_C - \bar{Y}_T = 6.8 - 7.0 = \boxed{-0.2}$$

WHY X_3 flags C, and effect coding measures against the grand mean.

(c) $H_0 : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4$. (d) $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$.

(e) $\beta_3 = \mu_D - \mu_A$, so $H_0 : \mu_D = \mu_A$.

D3

[4 MARKS] · TEST 2 Q4

Balanced, $n_j = 8$; means 8, 12, 13; **standard deviations** 2, 3, 1. $\psi_1 = \mu_1 - \mu_2$, $\psi_2 = \mu_1 - 2\mu_2 + \mu_3$.

(a) Orthogonal? (b) Find MSE. (c) Test $H_0 : \psi_2 = 0$ at 5%. (d) Build an **81%** simultaneous CI for ψ_2 using Sidak-Bonferroni, $k = 2$.

WHAT THIS QUESTION IS REALLY ASKING

Contrasts plus a Sidak interval. He gives standard deviations, so square them. The 81% confidence level is deliberately odd so you must actually use the formula rather than recall a t value.

SOLUTION D3

(a)

$$(1)(1) + (-1)(-2) + (0)(1) = 1 + 2 + 0 = 3 \neq 0 \Rightarrow \text{NO}$$

(b)

$$SSE = 7(2^2 + 3^2 + 1^2) = 7(14) = 98, \quad MSE = \frac{98}{24-3} = \frac{98}{21} = \boxed{4.6667}$$

WHY Standard deviations, so square them. $N = 24$, $a = 3$.

(c)

$$\hat{\psi}_2 = 8 - 2(12) + 13 = -3, \quad \sum \frac{c_j^2}{n_j} = \frac{1+4+1}{8} = 0.75$$

$$F = \frac{9}{4.6667(0.75)} = \frac{9}{3.5} = \boxed{2.5714}$$

$F_{0.05,1,21}$: table has 20 and 22 giving 4.35 and 4.30, so use either (or 4.325). $2.5714 < 4.30 \rightarrow \text{FTR}$.

WHY He announced this rule himself: use the exact df when listed, otherwise a neighbour, and **say which you used**.

(d) 81% confidence $\rightarrow \alpha = 0.19$.

$$\alpha_{SB} = 1 - (0.81)^{1/2} = 1 - 0.9 = 0.1, \quad t_{0.05,21} = 1.721$$

$$s_{\hat{\psi}_2} = \sqrt{4.6667(0.75)} = \sqrt{3.5} = 1.8708$$

$$-3 \pm 1.721(1.8708) = -3 \pm 3.2196 = \boxed{(-6.2196, 0.2196)}$$

WHY Halve α_{SB} again for the two-sided t lookup: 0.1 becomes 0.05. Zero is inside, matching the FTR in (c).

D4

[3 MARKS] · TEST 2 Q5

Balanced 2×3 factorial, $n = 2$, $N = 12$. $SSA = 18.75$, $SSB = 67.17$, $SSAB = 0.50$, $SSE = 4.50$. Test the interaction at 5%.

WHAT THIS QUESTION IS REALLY ASKING

A one-line interaction test. All the work is in the degrees of freedom.

SOLUTION D4

$$\text{df: AB} = (2-1)(3-1) = 2, \text{ Error} = N - ab = 12 - 6 = 6.$$

$$F = \frac{0.50/2}{4.50/6} = \frac{0.25}{0.75} = \boxed{0.3333}$$

$F_{0.05,2,6} = 5.14$, and $0.3333 < 5.14$, so **FTR**: no interaction.

WHY Main effects if you want them: $F_A = 18.75/0.75 = 25.0 > F_{0.05,1,6} = 5.99$ reject; $F_B = (67.17/2)/0.75 = 44.78 > 5.14$ reject.

D5

[5 MARKS] · TEST 1 Q5 AND TEST 2 Q6

(a) Show $\frac{SSE}{\sigma^2} \sim \chi^2(N-a)$ under H_0 .

(b) With $\hat{\psi}_1 = \sum_j c_j \bar{Y}_j$ and $\hat{\psi}_2 = \sum_j k_j \bar{Y}_j$, show $\text{Cov}(\hat{\psi}_1, \hat{\psi}_2) = \sigma^2 \sum_j \frac{c_j k_j}{n_j}$, then find the correlation.

WHAT THIS QUESTION IS REALLY ASKING

Two short proofs. Both are about naming facts rather than heavy algebra.

SOLUTION D5

(a)

$$\frac{SSE}{\sigma^2} = \frac{1}{\sigma^2} \sum_{j=1}^a (n_j - 1) S_j^2 = \sum_{j=1}^a \frac{(n_j - 1) S_j^2}{\sigma^2}$$

By the sampling theory of the normal distribution, $\frac{(n_j - 1) S_j^2}{\sigma^2} \sim \chi^2(n_j - 1)$.

By independence of the groups these are independent, and a sum of independent chi-squares is chi-square with the df added:

$$\sim \chi^2 \left(\sum_j (n_j - 1) \right) = \chi^2(N - a)$$

WHY Two named facts. Both are marked.

(b)

$$\text{Cov}(\hat{\psi}_1, \hat{\psi}_2) = \sum_j \sum_{\ell} c_j k_{\ell} \text{Cov}(\bar{Y}_j, \bar{Y}_{\ell}) \stackrel{\text{by independence}}{=} \sum_j c_j k_j V(\bar{Y}_j) = \sigma^2 \sum_j \frac{c_j k_j}{n_j}$$

WHY Use two different letters for the two sums. Off-diagonal terms die by independence; on the diagonal covariance becomes variance.

$$\text{Corr} = \frac{\sum_j c_j k_j / n_j}{\sqrt{\sum_j c_j^2 / n_j} \sqrt{\sum_j k_j^2 / n_j}}$$

WHY The σ^2 cancels. **The numerator is exactly the orthogonality condition**, so the correlation is zero precisely when the contrasts are orthogonal. That is what "orthogonal contrasts carry independent information" really means.

EXTRA DRILLS • UNITS NONE OF THE FOUR PAPERS TESTED

The gaps

The three papers above never touched RCB efficiency, Latin square, Kruskal-Wallis or power. Your final has 7 to 8 questions rather than 6, so these are where the extra questions most likely come from.

E1

RCB AND RELATIVE EFFICIENCY

Three treatments in four blocks, one observation per cell.

	BLK1	BLK2	BLK3	BLK4	TOTAL
A	20	24	17	19	80
B	25	27	21	23	96
C	26	31	27	24	108
Total	71	82	65	66	284

(a) ANOVA table. (b) Test treatments at 5%. (c) Compute and interpret the relative efficiency.

WHAT THIS QUESTION IS REALLY ASKING

A blocked design where the real question is the last part: **was blocking worth it?** The F test on blocks hints at it, but relative efficiency answers it properly, and it needs both degrees of freedom.

SOLUTION E1

BLOCKING, AND THE CF SHORTCUT, ONCE

A **nuisance** factor affects the response but you do not care about it. Left alone it inflates MSE and buries the treatment effect. **Blocking** gives it its own row so that variability leaves the error term. RCB is literally the two-way **additive** model; the only difference is intent.

Correction factor: $CF = (\text{grand total})^2 / N$. Then every SS is "sum of squared group totals, divided by how many observations made each total, minus CF".

Faster than the deviation form whenever you are given raw data.

$$CF = \frac{284^2}{12} = 6721.3333, \quad \sum y^2 = 6892$$

$$SST = 6892 - 6721.3333 = 170.6667$$

$$SS_{\text{Treat}} = \frac{80^2 + 96^2 + 108^2}{4} - CF = 6820 - 6721.3333 = 98.6667$$

$$SS_{\text{Block}} = \frac{71^2 + 82^2 + 65^2 + 66^2}{3} - CF = 6782 - 6721.3333 = 60.6667$$

$$SSE = 170.6667 - 98.6667 - 60.6667 = 11.3333$$

WHY Divide by **4** for treatment totals (each made from 4 numbers) and by **3** for block totals (each made from 3). SSE is the leftover.

SOURCE	DF	SS	MS	F
Treatment	2	98.6667	49.3333	26.1176
Block	3	60.6667	20.2222	10.7059
Error	6	11.3333	1.8889	
Total	11	170.6667		

(b) $H_0: \alpha_1 = \alpha_2 = \alpha_3 = 0$. $F = 26.1176 > F_{0.05,2,6} = 5.14 \rightarrow$ **reject**.

WHY Error df = $N - a - b + 1 = 12 - 3 - 4 + 1 = 6$.

(c) $df_{RCB} = 6$, $df_{CR} = N - a = 9$.

$$\widehat{RE} = \frac{(6+1)(9+3)}{(6+3)(9+1)} \cdot \frac{(b-1)MS_B + b(a-1)MSE}{(ab-1)MSE}$$

$$= \frac{84}{90} \cdot \frac{3(20.2222) + 4(2)(1.8889)}{11(1.8889)} = 0.9333 \times \frac{75.7778}{20.7778} = \boxed{3.4040}$$

WHY $RE = 3.40 > 1$, so blocking was **effective**: a completely randomised design would have needed about 3.4 times as many units for the same precision. $RE < 1$ would mean blocking cost more in df than it saved.

E2

LATIN SQUARE

Four assembly methods A, B, C, D. Rows are order of assembly (fatigue), columns are operator.

ORDER	OP1	OP2	OP3	OP4
1	C=10	D=14	A=7	B=8
2	B=7	C=18	D=11	A=8
3	A=5	B=10	C=11	D=9
4	D=10	A=10	B=12	C=14

Analyse fully and draw conclusions.

WHAT THIS QUESTION IS REALLY ASKING

A Latin square is just an ANOVA table with two extra rows. The only genuinely new things are hunting the treatment totals (the letters are scattered) and remembering error df is $(a-1)(a-2)$.

SOLUTION E2

LATIN SQUARE, ONCE

RCB removes **one** nuisance direction; a Latin square removes **two**. Each treatment appears exactly once in every row and once in every column, and that balance is what lets three effects be separated from only a^2 runs. Completely additive: no interactions. Error df is $(a-1)(a-2)$, from $(a^2-1) - 3(a-1)$.

Treatment totals (hunt each letter): A = 7 + 8 + 5 + 10 = 30, B = 8 + 7 + 10 + 12 = 37, C = 10 + 18 + 11 + 14 = 53, D = 14 + 11 + 9 + 10 = 44.

Row totals: 39, 44, 35, 46. Column totals: 32, 52, 41, 39. Grand total 164, $N = 16$.

WHY All three sets must add to 164. That is your check before doing any squaring.

$$CF = \frac{164^2}{16} = 1681, \quad \sum y^2 = 1834, \quad SST = 153$$

$$SS_{\text{Treat}} = \frac{30^2 + 37^2 + 53^2 + 44^2}{4} - 1681 = 1753.5 - 1681 = 72.5$$

$$SS_{\text{Row}} = \frac{39^2 + 44^2 + 35^2 + 46^2}{4} - 1681 = 1699.5 - 1681 = 18.5$$

$$SS_{\text{Col}} = \frac{32^2 + 52^2 + 41^2 + 39^2}{4} - 1681 = 1732.5 - 1681 = 51.5$$

$$SSE = 153 - 72.5 - 18.5 - 51.5 = 10.5$$

WHY Every total is built from 4 numbers, so every divisor is 4. **SSE has no formula of its own: it is the leftover.**

SOURCE	DF	SS	MS	F
Method	3	72.5	24.1667	13.8095
Order (rows)	3	18.5	6.1667	3.5238
Operator (cols)	3	51.5	17.1667	9.8095
Error	6	10.5	1.75	
Total	15	153		

All three against $F_{0.05,3,6} = 4.76$:

Method $13.8095 > 4.76 \rightarrow$ reject; assembly method affects assembly time.

Order $3.5238 < 4.76 \rightarrow$ FTR; the fatigue trend was not significant.

Operator $9.8095 > 4.76 \rightarrow$ reject; blocking on operator was worthwhile.

WHY Error df is 6, not 9. Row and column tests are design diagnostics, like the block F in RCB, not research conclusions.

E3

BIBD EXISTENCE

Verify that a BIBD with $a = 8$, $r = 8$, $k = 4$, $b = 16$ does not exist.

WHAT THIS QUESTION IS REALLY ASKING

A two-line existence proof. Compute lambda; if it is not a whole number, no such design exists. They design these so that r comes out clean and only lambda fails.

SOLUTION E3

Check the counting relation first: $ar = 8(8) = 64$ and $bk = 16(4) = 64$. Consistent, so no contradiction yet.

$$\lambda = \frac{r(k-1)}{a-1} = \frac{8(3)}{7} = \frac{24}{7} = 3.4286$$

WHY Not an integer. Since λ counts the number of blocks containing a given **pair** of treatments, it must be a whole number. Hence **no such design exists. Always check λ , not just r :** they design these so that r comes out clean and only λ fails.

E4

KRUSKAL-WALLIS

Normality is doubtful. Three groups, $n_j = 5$ each.

A: 12, 15, 9, 11, 14. B: 20, 18, 25, 22, 19. C: 16, 13, 17, 21, 10.

Test at 5%. State the hypotheses, statistic, critical value and decision.

WHAT THIS QUESTION IS REALLY ASKING

A rank test. The values themselves stop mattering once you rank them. Watch the degrees of freedom: it is $a-1$ and the distribution is chi-square, not F.

SOLUTION E4

WHY RANKS, ONCE

Replace the numbers by their **ranks** in the pooled sample and the shape of the distribution stops mattering. Use it when **normality or equal variance** fail. It does **not** rescue dependence. df is $a - 1$ and the distribution is chi-square, upper tail.

$N = 15$. Sorted: 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 25 with ranks 1 to 15.

A ranks 4, 7, 1, 3, 6 $\rightarrow R_A = 21$

B ranks 12, 10, 15, 14, 11 $\rightarrow R_B = 62$

C ranks 8, 5, 9, 13, 2 $\rightarrow R_C = 37$

WHY Rank the **pooled** list, then send each rank back to its group. Ties take the average rank. Check $21 + 62 + 37 = 120 = \frac{15(16)}{2}$.

$$H = \frac{12}{N(N+1)} \sum \frac{R_j^2}{n_j} - 3(N+1) = \frac{12}{240} \cdot \frac{21^2 + 62^2 + 37^2}{5} - 48$$

$$= 0.05 \left(\frac{5654}{5} \right) - 48 = 56.54 - 48 = \boxed{8.54}$$

$H_0: \mu_1 = \mu_2 = \mu_3$. $\chi_{0.05,2}^2 = 5.991$, and $8.54 > 5.991$, so **reject**.

WHY df is $a - 1 = 2$, not $N - a$.

E5

EFFECT SIZE AND POWER

(a) From an ANOVA table, $SSA = 6.30$ and $SSE = 11.20$: find and classify R^2 . (b) Two groups, means 8 and 12, $n_1 = n_2 = 8$, sds 2 and 3: find and classify d . (c) Name three ways to increase power.

WHAT THIS QUESTION IS REALLY ASKING

Effect size is about **how big**, not **how significant**. The two are different questions and he tests that you know it.

SOLUTION E5

$$R^2 = \frac{6.30}{17.50} = 0.36 \Rightarrow \text{large}$$

$$S_{12}^2 = \frac{7(2^2) + 7(3^2)}{8 + 8 - 2} = \frac{28 + 63}{14} = 6.5, \quad S_{12} = 2.5495$$

$$d = \frac{8 - 12}{2.5495} = -1.5689, \quad |d| > 0.8 \Rightarrow \text{large}$$

WHY S_{12} is the pooled sd, the same one as in the two-sample t test.

(c) Increase n ; increase α ; reduce the error variance by blocking or by adding a covariate. (A larger true effect also raises power but is not under your control.)

THE IDEA HE TESTS

Statistical significance is not practical significance. With a huge sample, a trivial difference becomes significant. If a question says "very large sample, very small p-value, tiny effect size", the answer is **statistically significant but the practical importance may be limited**.

VARIANTS • SAME TOPICS, THE OTHER WAYS HE CAN ASK THEM

Coverage of what the past papers missed

Everything above is a question he has actually set. This section covers the **variants**: material that is examinable, sits on the formula sheet or in the lectures, but has not appeared in the four papers. With 7 to 8 questions this year rather than 6, these are live.

V1

TUKEY HSD • ON THE SHEET, NEVER YET TESTED

Using the C1 data (means 62, 66, 52; $n_j = 5$; $MSE = 28.3333$; $a = 3$, $N - a = 12$), and given $q_{0.05, 3, 12} = 3.77$:

(a) Compute the Tukey HSD critical difference. (b) Which pairs of treatments differ? (c) Give a 95% Tukey CI for $\mu_1 - \mu_3$.

WHAT THIS QUESTION IS REALLY ASKING

Tukey is the pairwise specialist. One critical difference is computed once (balanced design) and then compared against every observed gap.

SOLUTION V1

TUKEY, ONCE

Tukey controls the family-wise rate across **all pairwise** comparisons at once, using the **studentized range** distribution q rather than t . It is the **most powerful** option when you want every pair and the design is balanced. It does **not** handle general contrasts; for those you need Bonferroni or Scheffe.

$$HSD = \frac{q_{\alpha, a, N-a}}{\sqrt{2}} \sqrt{MSE \left(\frac{1}{n_j} + \frac{1}{n_k} \right)} = \frac{3.77}{\sqrt{2}} \sqrt{28.3333 \left(\frac{1}{5} + \frac{1}{5} \right)}$$

$$= 2.6658 \sqrt{28.3333(0.4)} = 2.6658 \sqrt{11.3333} = 2.6658(3.3665) = \boxed{8.9744}$$

WHY The q subscripts are α , then a (number of groups, **not** $a - 1$), then error df. Balanced here so every pair has the same HSD; unbalanced would need recomputing per pair.

(b) Compare each $|\bar{Y}_j - \bar{Y}_k|$ to 8.9744: $|62 - 66| = 4 \rightarrow$ not significant; $|62 - 52| = 10 \rightarrow$ **significant**; $|66 - 52| = 14 \rightarrow$ **significant**. Treatment 3 differs from both 1 and 2; treatments 1 and 2 do not differ.

WHY "If the observed difference exceeds the HSD value, the means differ." Report which pairs, in words.

(c)

$$(\bar{Y}_1 - \bar{Y}_3) \pm HSD = 10 \pm 8.9744 = \boxed{(1.0256, 18.9744)}$$

WHY Zero is outside, consistent with (b). The Tukey CI is just the difference plus or minus the same HSD.

V2

SCHEFFE

Same data. After looking at the data you decide to test $\psi_1 = \mu_1 + \mu_2 - 2\mu_3$ ($\hat{\psi}_1 = 24$, $s_{\hat{\psi}_1} = 5.8310$). Use Scheffe at 5%. Why is Scheffe the right choice here?

WHAT THIS QUESTION IS REALLY ASKING

The point is not the arithmetic, it is **why Scheffe**: the contrast was chosen after seeing the data, which rules out Bonferroni, Sidak and Tukey.

SOLUTION V2

SCHEFFE, ONCE

Scheffe controls the family-wise rate over **every possible contrast simultaneously**, including ones you invented after seeing the data. That is an infinite family, so it is the **most conservative** and least powerful. It is the only honest option for unplanned contrasts.

Scheffe critical value:

$$S = \sqrt{(a-1)F_{\alpha, a-1, N-a}} = \sqrt{2 \times 3.89} = \sqrt{7.78} = 2.7893$$

Test statistic:

$$|t_{\text{obs}}| = \frac{|\hat{\psi}_1|}{s_{\hat{\psi}_1}} = \frac{24}{5.8310} = 4.1160$$

Since $4.1160 > 2.7893$, **reject**: the contrast is significant even by Scheffe.

WHY Scheffe replaces the usual $t_{\alpha/2, N-a}$ multiplier with $\sqrt{(a-1)F}$. Compare: the Bonferroni multiplier in C2 was 2.179 and Scheffe's is 2.7893, visibly harsher. That is the price of not planning ahead.

Why Scheffe: the contrast was chosen **after** seeing the data. Bonferroni and Sidak require the family to be fixed in advance, and Tukey only covers pairwise differences. Scheffe is the only procedure valid for a contrast you did not pre-specify.

V3

HELMERT CONTRASTS

For $a = 3$ treatments, construct a set of Helmert contrasts and verify they are mutually orthogonal (balanced design).

WHAT THIS QUESTION IS REALLY ASKING

A construction question. Follow the pattern and orthogonality is automatic; verifying it is one dot product.

SOLUTION V3

HELMERT CONSTRUCTION, ONCE

Compare group 1 to the average of all the rest. Then zero out group 1 and compare group 2 to the average of those after it. Continue. Formally: $c_1 = 1$ and $c_j = -\frac{1}{a-1}$ for $j > 1$; then for contrast j , set $c_k = 0$ for $k \leq j-1$, $c_j = 1$, $c_k = -\frac{1}{a-j}$ for $k \geq j+1$. It always produces $a-1$ mutually orthogonal contrasts.

$$\psi_1 : \mathbf{c}_1 = (1, -\frac{1}{2}, -\frac{1}{2}) \quad \psi_2 : \mathbf{c}_2 = (0, 1, -1)$$

Both sum to zero, so both are contrasts.

$$\mathbf{c}_1 \cdot \mathbf{c}_2 = (1)(0) + (-\frac{1}{2})(1) + (-\frac{1}{2})(-1) = 0 - \frac{1}{2} + \frac{1}{2} = 0$$

Orthogonal.

WHY Only $a-1 = 2$ contrasts, which is the maximum size of a mutually orthogonal set. Their sums of squares add to SSA exactly.

V4

UNIT 2 • TWO-SAMPLE T, POOLED VS WELCH

Two groups. $n_1 = 10$, $\bar{Y}_1 = 25.4$, $s_1 = 3.2$; $n_2 = 12$, $\bar{Y}_2 = 22.1$, $s_2 = 3.6$. A test of equal variances gives $p = 0.62$.

(a) Which t test should you use, and why? (b) Carry it out at 5%. (c) What is the relationship between this test and a one-way ANOVA on the same two groups?

WHAT THIS QUESTION IS REALLY ASKING

Choosing between pooled and Welch, and justifying the choice. The final part is the conceptual payoff: for two groups, the t test and ANOVA are the same test.

SOLUTION V4

(a) The variance test gives $p = 0.62 > 0.05$, so there is no evidence against equal variances. Use the **pooled** t test with $df = n_1 + n_2 - 2 = 20$.

WHY Test the variances first, then choose. If they had differed, or the evidence were inconclusive, you would use **Welch** with the fractional df from the sheet. Stating *which* and *why* is where the mark is.

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2} = \frac{9(10.24) + 11(12.96)}{20} = \frac{92.16 + 142.56}{20} = 11.736$$

$$t = \frac{25.4 - 22.1}{\sqrt{11.736 \left(\frac{1}{10} + \frac{1}{12} \right)}} = \frac{3.3}{\sqrt{11.736(0.18333)}} = \frac{3.3}{1.4668} = \boxed{2.2498}$$

$$t_{0.025, 20} = 2.086. \text{ Since } 2.2498 > 2.086, \text{ reject } H_0 : \mu_1 = \mu_2.$$

WHY Two-sided, so $\alpha/2 = 0.025$. Note s_p^2 is exactly MSE for two groups.

(c) They are equivalent: with $a = 2$, $F = t^2$. Here $2.2498^2 = 5.0616$, which is what a one-way ANOVA would report as $F_{1,20}$, and $F_{0.05, 1, 20} = 4.35 = 2.086^2$. Same decision.

V5

UNIT 5 • CHECKING ASSUMPTIONS FROM OUTPUT

A one-way ANOVA on 4 groups produces: a normal Q-Q plot of residuals with points falling close to the line; a residuals-versus-fitted plot showing a clear funnel shape widening to the right; and the data were collected by measuring the same 10 subjects under all four conditions.

(a) State the three assumptions and assess each. (b) What should the analyst do?

WHAT THIS QUESTION IS REALLY ASKING

A diagnosis question with no numbers. Match each piece of evidence to an assumption, then rank the damage. The trap is reaching for Kruskal-Wallis when the broken assumption is independence, which it cannot fix.

SOLUTION V5

ASSUMPTION	EVIDENCE	VERDICT
Independence	Same 10 subjects measured under all four conditions	Violated. These are repeated measurements on the same units, so observations are not independent.
Equal variance	Funnel in residuals vs fitted	Violated. Spread grows with the fitted value.
Normality	Q-Q points near the line	Satisfied.

SEVERITY RANKING, ONCE

Independence is the most serious: no correction rescues it, and the course offers no fix beyond using a model with correlated errors. **Equal variance** is next. **Normality** is the least serious, because t and F are robust to it except in small samples.

(b) The independence violation is fatal for this model, so the first move is a design-level fix: the repeated measurements mean this is **not** a completely randomised design, and a model that accounts for the subject (for example treating subject as a block, which is the RCB structure) is required. Kruskal-Wallis would **not** help, because it also assumes independence. Had independence held and only equal variance failed, Kruskal-Wallis would then be the right alternative.

THE TRAP

Students reach for Kruskal-Wallis whenever an assumption fails. It only covers normality and equal variance. Repeated measurements on the same unit are "repeated measurements", not replicates, and Unit 1 says so explicitly.

V6

ANCOVA WITH TWO COVARIATES · PRACTICE 9 Q1

Four diabetes medicines: Alpha-glucosidase (A), Dopamine agonist (D), Insulin (I), Placebo (P). Life expectancy y . Covariates: age x_1 and blood pressure x_2 . Placebo is the reference.

(a) Write the parallel planes model. (b) Write H_0 for: (i) controlling for age, there is no blood pressure effect; (ii) among patients of the same age and blood pressure, the average life expectancy on A and D equals that on I and P; (iii) controlling for both, all medications give the same mean.

(c) Write the model allowing both covariate effects to vary by medication. (d) Write H_0 for: (i) equal slopes for all medications; (ii) controlling for both covariates, medication has no significant effect.

WHAT THIS QUESTION IS REALLY ASKING

ANCOVA with two covariates. Nothing conceptually new, but the interaction model now needs **six** extra terms, not three, because each covariate interacts with each indicator.

SOLUTION V6

(a) Let I_A, I_D, I_I be indicators for medicines A, D and I (Placebo is the reference and gets none).

$$E[Y|X_1 = x_1, X_2 = x_2] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 I_A + \beta_4 I_D + \beta_5 I_I$$

WHY Two covariates means two slope terms, β_1 and β_2 . Three indicators for four groups. Nothing else changes.

MEDICINE	$E[Y X_1, X_2]$
A	$(\beta_0 + \beta_3) + \beta_1 x_1 + \beta_2 x_2$
D	$(\beta_0 + \beta_4) + \beta_1 x_1 + \beta_2 x_2$
I	$(\beta_0 + \beta_5) + \beta_1 x_1 + \beta_2 x_2$
P (ref)	$\beta_0 + \beta_1 x_1 + \beta_2 x_2$

(b)(i) $H_0 : \beta_2 = 0$

WHY β_2 is the only place blood pressure appears.

(ii) Average of A and D equals average of I and P:

$$\frac{(\beta_0 + \beta_3) + (\beta_0 + \beta_4)}{2} = \frac{(\beta_0 + \beta_5) + \beta_0}{2}$$

$$\Rightarrow \frac{\beta_3 + \beta_4}{2} = \frac{\beta_5}{2} \Rightarrow H_0 : \beta_3 + \beta_4 = \beta_5$$

WHY Average the two rows on each side, cancel the common $\beta_0 + \beta_1 x_1 + \beta_2 x_2$. Placebo contributes nothing because it is the reference. Pure algebra off the table.

(iii) $H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

WHY All four rows collapse to the reference row.

(c) Let each covariate interact with each indicator:

$$E[Y|x_1, x_2] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 I_A + \beta_4 I_D + \beta_5 I_I \\ + \beta_6 x_1 I_A + \beta_7 x_1 I_D + \beta_8 x_1 I_I + \beta_9 x_2 I_A + \beta_{10} x_2 I_D + \beta_{11} x_2 I_I$$

WHY Two covariates times three indicators gives **six** interaction terms, not three. This is the only thing that changes when a second covariate appears.

(d)(i) $H_0 : \beta_6 = \beta_7 = \beta_8 = \beta_9 = \beta_{10} = \beta_{11} = 0$

(ii) $H_0 : \beta_3 = \beta_4 = \beta_5 = \beta_6 = \dots = \beta_{11} = 0$

WHY Equal slopes means every interaction vanishes. "No medication effect at all" means the group rows collapse entirely onto the reference, so **both** the intercept shifts and every slope shift must be zero: all nine.

V7

PROOF VARIANT · EXPECTED MEAN SQUARES

(a) Show $E[MSE] = \sigma^2$. (b) State $E[MSA]$ and explain in one sentence why this makes the F test work.

WHAT THIS QUESTION IS REALLY ASKING

This is the theory behind the F test. Being able to state $E[MSA]$ explains in one line why a big F is evidence against the null.

SOLUTION V7

(a) Within group j , S_j^2 is unbiased for σ^2 , so $E[(n_j - 1)S_j^2] = (n_j - 1)\sigma^2$. Summing over the groups:

$$E[SSE] = \sum_{j=1}^a (n_j - 1)\sigma^2 = (N - a)\sigma^2 \\ \Rightarrow E[MSE] = E\left[\frac{SSE}{N - a}\right] = \frac{(N - a)\sigma^2}{N - a} = \sigma^2$$

WHY Nothing about the treatment means enters, which is exactly why MSE is unbiased for σ^2 **whether or not** H_0 is true. That is also the answer to the MSE multiple-choice question.

(b)

$$E[MSA] = \sigma^2 + \frac{\sum_{j=1}^a n_j \alpha_j^2}{a - 1}$$

WHY Under H_0 every $\alpha_j = 0$, the second term vanishes, and both mean squares estimate σ^2 , so $F \approx 1$. When H_0 is false the second term is strictly positive, so **only the numerator inflates** and F grows. Hence a large F is evidence against H_0 , and the test is **upper tail only**. The full derivation of (b) is Practice 5 Question 4.

V8

PROOF VARIANT · THE PARTITION

Show that $SST = SSA + SSE$.

WHAT THIS QUESTION IS REALLY ASKING

The partition proof. It is short and the entire trick is that deviations from a mean add to zero.

SOLUTION V8

Add and subtract the group mean inside the total deviation:

$$\sum_j \sum_i (Y_{ij} - \bar{Y}_T)^2 = \sum_j \sum_i [(\bar{Y}_j - \bar{Y}_T) + (Y_{ij} - \bar{Y}_j)]^2 \\ = \underbrace{\sum_j \sum_i (\bar{Y}_j - \bar{Y}_T)^2}_{SSA} + \underbrace{\sum_j \sum_i (Y_{ij} - \bar{Y}_j)^2}_{SSE} + 2 \sum_j (\bar{Y}_j - \bar{Y}_T) \sum_i (Y_{ij} - \bar{Y}_j)$$

WHY Expand the square of a sum. The first bracket has no i , so summing over i gives the factor n_j , which recovers $SSA = \sum n_j (\bar{Y}_j - \bar{Y}_T)^2$.

The cross term vanishes because, **by definition of a sample mean**,

$$\sum_{i=1}^{n_j} (Y_{ij} - \bar{Y}_j) = 0 \quad \text{for every } j$$

Hence $SST = SSA + SSE$.

WHY **The whole proof is "deviations from a mean sum to zero"**. Say that sentence explicitly when you kill the cross term.

V9

TWO-WAY WITH ONE OBSERVATION PER CELL

A two-factor study has $a = 4$, $b = 3$ and exactly **one** observation per cell. (a) Why can you not fit the interaction? (b) Write the model you must use and give the full df column. (c) How would you test factor A?

WHAT THIS QUESTION IS REALLY ASKING

A design-logic question. With one observation per cell there is no room left to estimate error, so you are forced to assume additivity.

SOLUTION V9

(a) $N = ab = 12$. If the interaction were fitted, error df would be $N - ab = 12 - 12 = 0$: there would be nothing left to estimate σ^2 with, so no F test is possible.

WHY With one observation per cell, the cell mean **is** the observation, so the model fits the data perfectly and leaves no residual.

(b) The **additive** model, $Y_{jk} = \mu + \alpha_j + \beta_k + \varepsilon_{jk}$, assuming no interaction.

SOURCE	DF
Factor A	$a - 1 = 3$
Factor B	$b - 1 = 2$
Error	$(a - 1)(b - 1) = 6$
Total	$N - 1 = 11$

WHY The interaction df becomes the error df. Check: $3 + 2 + 6 = 11$. This is the identical structure to RCB, which is why RCB is "two-way with no interaction".

(c) $F = MSA/MSE \sim F_{a-1,(a-1)(b-1)} = F_{3,6}$. Reject if it exceeds $F_{0.05,3,6} = 4.76$.

V10

UNIT 1 VOCABULARY, THE PARTS THAT OPEN EVERY QUESTION

A hospital compares three physiotherapy protocols. Six clinics are chosen; within each clinic, patients are randomly assigned to one of the three protocols. Recovery time in days is recorded. Each clinic uses its own physiotherapists.

(a) Experimental unit? (b) Factor and levels? (c) Response, and its type? (d) Is this an experiment or an observational study, and why? (e) Is "clinic" a treatment factor or a block? (f) Are physiotherapist and clinic crossed or nested?

WHAT THIS QUESTION IS REALLY ASKING

All the Unit 1 vocabulary in one place. The subtle parts are recognising a block rather than a factor, and spotting nesting when each group has its own personnel.

SOLUTION V10

UNIT 1 DEFINITIONS, ONCE

Experimental unit: the entity that receives a treatment independently of the others. **Factor:** a categorical variable being controlled; its values are **levels**.

Treatment: one level, or one combination of levels in a factorial design. **Experiment:** the researcher **assigns** the levels (supports causation). **Observational study:** the levels are merely **recorded** (association only). **Nested:** the levels of one factor are different within each level of the other. **Confounding:** a nuisance variable left unmanipulated, so its effect cannot be separated from the treatment.

(a) The **individual patient**, since protocols are assigned to patients.

(b) The factor is **physiotherapy protocol**, with 3 levels.

(c) **Recovery time in days**, which is **quantitative** (continuous).

(d) An **experiment**: patients are **randomly assigned** to protocols by the researcher, rather than choosing their own. This is what supports a causal conclusion.

(e) A **block**. Clinic is a nuisance source of variation, not a factor of interest, and every protocol appears within every clinic, so this is a randomized complete block design with $a = 3$, $b = 6$.

(f) **Nested**. Each clinic has its own physiotherapists, so "physiotherapist 1" at clinic A is a different person from "physiotherapist 1" at clinic B. They are not the same levels appearing across clinics, so no interaction between physiotherapist and clinic is estimable.

V11

INTERACTION PLOT READING

In a two-factor study, the interaction plot shows the line for level a_1 rising steeply from b_1 to b_3 while the line for a_2 is almost flat, and the two lines cross between b_1 and b_2 .

(a) What does the plot suggest? (b) If the interaction F test confirms it, what may you conclude about the main effects? (c) Describe the finding without using the word "interaction".

WHAT THIS QUESTION IS REALLY ASKING

Reading a picture. Crossing lines mean the ranking flips, so no single statement about one factor can be true on its own.

SOLUTION V11

(a) Non-parallel and crossing lines suggest a **substantial interaction**. Parallel lines would mean none.

(b) If the interaction is significant you must **stop** and not interpret the main effects on their own. A single statement such as "level a_1 is better" would be false, because which level is better **depends on the level of B**. Report the pattern instead.

(c) The effect of Factor A on the response is **not the same at every level of Factor B**. At b_1 level a_2 gives the higher response, while at b_3 level a_1 gives the higher response, so the better choice of A depends on which level of B you are in.

HE ASKS THIS IN WORDS ON ALMOST EVERY TWO-WAY QUESTION

Have the sentence ready: "the effect of [factor A] on [response] is not the same at every level of [factor B]", then name a concrete case from the data.

V12

BUILDING SST FROM RAW DATA

Three groups: A: 4, 6, 5. B: 9, 8, 10. C: 7, 5, 6. Build the complete ANOVA table from scratch and test at 5%.

WHAT THIS QUESTION IS REALLY ASKING

Building everything from raw numbers. Use the correction-factor route; it avoids the rounding trap that comes from computing the grand mean first.

SOLUTION V12

Group totals 15, 27, 18; grand total 60; $N = 9$, $a = 3$, $n_j = 3$.

$$CF = \frac{60^2}{9} = 400, \quad \sum y^2 = 16 + 36 + 25 + 81 + 64 + 100 + 49 + 25 + 36 = 432$$

$$SST = 432 - 400 = 32$$

$$SSA = \frac{15^2 + 27^2 + 18^2}{3} - 400 = \frac{225 + 729 + 324}{3} - 400 = 426 - 400 = 26$$

$$SSE = 32 - 26 = 6$$

WHY Divide each squared group total by $n_j = 3$, the number of observations in that total. This is the fastest route when raw data are given.

SOURCE	DF	SS	MS	F
Treatment	2	26	13	13
Error	6	6	1	
Total	8	32		

$F_{0.05,2,6} = 5.14$, and $13 > 5.14$, so **reject** $H_0 : \mu_A = \mu_B = \mu_C$.

$$R^2 = \frac{26}{32} = 0.8125$$

WHY Cross-check with the deviation form: means 5, 9, 6, grand mean 6, so $SSA = 3(1) + 3(9) + 3(0) = 30$. That disagrees, so recheck: means are $15/3 = 5$, $27/3 = 9$, $18/3 = 6$, grand mean $60/9 = 6.6667$. Then $SSA = 3(2.7779) + 3(5.4449) + 3(0.4449) = 26.0$. Agrees. **The grand mean is 6.6667, not 6**, and that is exactly the slip the computational form avoids.

V13

WELCH, AND THE DETAIL HE FLAGS ON THE SLIDE

Two groups: $n_1 = 8$, $s_1^2 = 4.0$; $n_2 = 10$, $s_2^2 = 25.0$. A variance test gives $p = 0.01$. Which test, and what degrees of freedom?

WHAT THIS QUESTION IS REALLY ASKING

One detail decides this: Welch's df rounds **down**. He put that on a slide precisely because it is easy to get backwards.

SOLUTION V13

The variance test rejects equality ($p = 0.01 < 0.05$), so use **Welch**, not pooled.

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{(0.5 + 2.5)^2}{\frac{0.25}{7} + \frac{6.25}{9}} = \frac{9}{0.0357 + 0.6944} = \frac{9}{0.7301} = 12.3269$$

WHY Satterthwaite approximation, on the formula sheet.

Round down to 12. Use $t_{0.025,12} = 2.179$.

WHY His Unit 2 slide says it explicitly: "Note: may not be an integer, round down to the nearest integer." Rounding up would be wrong and it is the kind of one-line detail he puts on a slide precisely because he intends to test it.

ALSO FROM HIS SLIDES

Welch is used when the variances are unequal **or the evidence about equality is inconclusive**. Its assumptions are only: independent random samples, and approximate normality. It does **not** assume equal variances, which is the whole point.

V14

TWO "WHY?" QUESTIONS HE WRITES ON HIS OWN SLIDES

- Why does every contrast have $df = 1$?
- Why may a contrast be tested equivalently with a t test rather than an F test?
- Derive $\alpha_{SB} = 1 - (1 - \alpha_{FW})^{1/k}$.

WHAT THIS QUESTION IS REALLY ASKING

Three short 'why' questions he has literally written on his own slides. Each is one or two lines.

SOLUTION V14

(a) His rule on the slide: **df equals the number of equal signs in your system of tests**. A contrast states one equation, $\psi = 0$, so one equal sign, so $df = 1$. Compare the global test $\mu_1 = \mu_2 = \dots = \mu_a$, which has $a - 1$ equal signs and therefore $a - 1$ df. This rule is worth remembering because it also explains why the interaction has $(a - 1)(b - 1)$ df.

(b) Because $t^2 = F$ when the numerator has 1 df. Specifically $t_{\text{obs}} = \hat{\psi}/s_{\hat{\psi}} \sim t_{N-a}$ and squaring it gives $F_{1, N-a}$. The two tests give identical decisions; the t version is more convenient if you also want a confidence interval or a one-sided test.

(c) If the k tests are independent and each uses level α_{SB} , then each avoids a false positive with probability $1 - \alpha_{SB}$, so

$$P(\text{no false positive anywhere}) = (1 - \alpha_{SB})^k$$

$$\alpha_{FW} = 1 - (1 - \alpha_{SB})^k \implies (1 - \alpha_{SB})^k = 1 - \alpha_{FW}$$

$$\implies 1 - \alpha_{SB} = (1 - \alpha_{FW})^{1/k} \implies \boxed{\alpha_{SB} = 1 - (1 - \alpha_{FW})^{1/k}}$$

WHY He writes "(why?)" beside this on the slide, and W2025 Test 2 asked for the derivation outright. Two lines, worth easy marks.

V15

EFFECT ESTIMATES AND THE ZERO-SUM CHECK

A two-factor study gives row means 61.25, 60.00, 70.75, 65.50 and column means 63.00, 65.75, with grand mean 64.375. Find all $\hat{\alpha}_j$ and $\hat{\beta}_k$, and state the check.

WHAT THIS QUESTION IS REALLY ASKING

A free arithmetic check dressed as a question. Effects always sum to zero within each factor.

SOLUTION V15

$$\hat{\alpha}_j = \bar{Y}_{Aj} - \bar{Y}_T : \quad -3.125, -4.375, 6.375, 1.125$$

$$\hat{\beta}_k = \bar{Y}_{Bk} - \bar{Y}_T : \quad -1.375, 1.375$$

WHY An effect is always "this margin's mean minus the grand mean".

Check: $-3.125 - 4.375 + 6.375 + 1.125 = 0$ and $-1.375 + 1.375 = 0$.

WHY His slide states it in capitals: the estimated effects **sum to ZERO**, for each factor separately. It is a free arithmetic check, and it also mirrors the model constraint $\sum \alpha_j = 0$ used in every proof.

V16

POWER AND SAMPLE SIZE FROM R OUTPUT

An investigator plans a balanced one-way study with 4 groups.

```
pwr.anova.test(k = 4, n = 4, f = 0.5, sig.level = 0.05, power = NULL)
      k = 4      n = 4      f = 0.5      sig.level = 0.05      power = 0.2685639
NOTE: n is number in each group

pwr.anova.test(k = 4, n = NULL, f = 0.5, sig.level = 0.05, power = 0.8)
      n = 11.92613

pwr.anova.test(k = 4, n = NULL, f = 0.25, sig.level = 0.05, power = 0.8)
      n = 44.5993
```

(a) Interpret the first output. (b) How many subjects in total are needed for 80% power at $f = 0.5$? (c) Why does the required n jump when f drops to 0.25?

WHAT THIS QUESTION IS REALLY ASKING

Reading power output rather than computing it. Two details: n is **per group**, and required sample sizes round **up**, the opposite of Welch df.

SOLUTION V16

(a) With 4 groups of 4 subjects each and a large effect size ($f = 0.5$) at $\alpha = 0.05$, the power is only **0.2686**. So there is roughly a 27% chance of detecting a real effect of that size, and a 73% chance of a Type II error. The study is badly underpowered.

(b) $n = 11.93$ **per group**. Sample size must be a whole number and you cannot have less power than required, so **round up to 12 per group**, giving $4 \times 12 = 48$ subjects in total.

TWO DETAILS HE FLAGS

n is the number in each group, written in capitals on all three of his slides, so multiply by k for the total. And unlike the Welch df (round **down**), a required sample size rounds **up**, because rounding down would leave you under the target power.

(c) Power depends on effect size, and a smaller effect is harder to detect. Halving f from 0.5 to 0.25 raises the required n per group from about 12 to about 45, roughly a fourfold increase, because the required sample size grows approximately with $1/f^2$. This is the practical reason small effects need very large studies, and it is the same point as "statistical significance is not practical significance" seen from the other side.

V17

WHICH PROCEDURE? A PURE JUDGEMENT QUESTION

For each scenario say which multiple-comparison procedure is most appropriate and why.

- (a) A balanced design with 5 groups; the researcher wants every pairwise difference.
- (b) Three specific contrasts written into the grant proposal before data collection.
- (c) After plotting the data the researcher notices groups 1 and 4 look high and wants to test $\mu_1 + \mu_4 - \mu_2 - \mu_3$.
- (d) A single pre-planned contrast, and nothing else.

WHAT THIS QUESTION IS REALLY ASKING

Pure judgement, no numbers. Match the scenario to the procedure and justify in one line each.

SOLUTION V17

	PROCEDURE	WHY
(a))	Tukey HSD	All pairwise differences in a balanced design is exactly Tukey's case, and it is the most powerful there.
(b))	Sidak-Bonferroni (Bonferroni acceptable)	A fixed family of $k = 3$ comparisons specified in advance. Sidak is slightly more powerful than Bonferroni and is otherwise identical to apply.
(c))	Scheffe	The contrast was chosen after seeing the data. Bonferroni and Sidak require the family fixed beforehand, and Tukey covers only pairwise. Scheffe is the only honest option.
(d))	No correction	With a single test there is no family, so $k = 1$ and nothing is inflated. He gave this exact hint on Test 2.

CONSERVATISM ORDER, ONCE

Tukey (most powerful, least conservative) → Bonferroni → Scheffe (least powerful, most conservative). Scheffe pays for covering infinitely many contrasts.

BOTH OF HIS TESTS OPENED WITH THESE • 2 MARKS, 0.5 EACH

Multiple choice

The first eight are **his actual questions with his actual answers**. The rest are the same ideas in his style.

MC 1 TO 8 • REAL

- 1. Role of randomization?** (A) guarantees nuisance variables eliminated (B) spreads uncontrolled nuisance variables across treatment groups (C) only for observational studies (D) removes need for replication
- 2. Nuisance variable that cannot be held constant, best strategy?** (A) block, grouping similar units before assigning (B) treat as error, no adjustment (C) several measurements per unit, averaged (D) remove affected units
- 3. MSE in one-way ANOVA?** (A) unbiased for σ^2 only under H_0 (B) unbiased whether or not H_0 holds (C) estimates variation between treatment means (D) grows mainly when the means differ
- 4. MSA?** (A) always estimates σ^2 (B) estimates only within-treatment variability (C) always smaller than MSE when H_0 is false (D) under H_0 estimates σ^2 , otherwise tends to be larger
- 5. Four contrasts chosen before seeing the data, and only those four are of interest. Which procedure?** (A) Tukey (B) Bartlett (C) Scheffe (D) Bonferroni
- 6. Shapiro-Wilk p below 0.05, Bartlett p above 0.05?** (A) both questionable (B) observations dependent (C) equal variance questionable, normality fine (D) normality questionable, no evidence against equal variances
- 7. Huge sample, tiny p-value, near-zero effect size?** (A) the effect must be important (B) H_0 is certainly false everywhere (C) statistically significant, but practical importance may be limited (D) a small effect means the assumptions are violated
- 8. Two contrasts share some treatment means. Which is correct?** (A) they may still be orthogonal, depending on coefficients and sample sizes (B) they cannot be orthogonal (C) orthogonal only if both significant (D) automatically orthogonal if the coefficients sum to zero

ANSWERS 1 TO 8

- 1. B.** It does not remove nuisance variables, it stops them lining up with treatment.
- 2. A.** Definition of blocking. Option B is the mistake blocking exists to fix.
- 3. B.** MSE only looks inside groups, so it never sees the treatment means. See V7(a).
- 4. D.** $E[MSA] = \sigma^2 + \frac{\sum n_i \alpha_i^2}{a-1}$. See V7(b).
- 5. D.** A fixed pre-planned family is Bonferroni's case.
- 6. D.** Shapiro-Wilk tests normality (rejected); Bartlett tests equal variance (not rejected).
- 7. C.** Significance is not importance.
- 8. A.** Orthogonality is $\sum c_{1j}c_{2j}/n_j = 0$; sharing groups is irrelevant.

MC 9 TO 20 • SAME STYLE

9. **Interaction is significant. Do what?** (A) interpret main effects as usual (B) do not interpret main effects alone; describe how one factor's effect changes across the other (C) refit without interaction (D) use Kruskal-Wallis
10. **Block F not significant in an RCB?** (A) treatment test invalid (B) blocking removed little variability, a CR design would have been about as good (C) re-randomise (D) treatments do not differ
11. **Most serious assumption violation?** (A) normality (B) equal variance (C) independence (D) equally serious
12. **Latin square with $a = 5$, error df?** (A) 16 (B) 12 (C) 20 (D) 24
13. **BIBD with $a = 6, k = 3, r = 5$?** (A) exists with $\lambda = 2$ (B) impossible, λ not an integer (C) impossible since $k > a$ (D) $b = 9$
14. **H_0 of equal means in the cell means model?** (A) $\gamma_1 = \dots = \gamma_{a-1} = 0$ (B) $\gamma_1 = \dots = \gamma_a$ (C) $\gamma_0 = 0$ (D) $\sum \gamma_j = 0$
15. **Homogeneity of regression slopes means** (A) equal group means (B) no treatment-by-covariate interaction (C) the covariate has no effect (D) constant error variance
16. **Kruskal-Wallis is preferred when** (A) observations are dependent (B) normality or equal variance is violated (C) you want more power (D) sample sizes are unequal
17. **In effect coding the intercept estimates** (A) the reference group mean (B) the grand mean (C) zero (D) the largest group mean
18. **Welch's degrees of freedom, when not a whole number, should be** (A) rounded up (B) rounded down (C) rounded to nearest (D) left as a decimal
19. **Every contrast has how many degrees of freedom?** (A) $a - 1$ (B) $N - a$ (C) 1 (D) depends on the coefficients
20. **A required sample size from a power calculation comes out as 11.93 per group. You should use** (A) 11 (B) 12 (C) 11.93 (D) 48 per group
21. **Measuring the same subject at three time points gives** (A) three replicates (B) repeated measurements, which are not independent (C) a Latin square (D) three blocks

ANSWERS 9 TO 20

9. **B.** "It depends", so a single main-effect statement misleads.
10. **B.** The block F is a design diagnostic, never a research conclusion. Confirm with RE.
11. **C.** Independence. No fix in this course. Normality is least serious.
12. **B.** $(a-1)(a-2) = 4 \times 3 = 12$.
13. **A.** $\lambda = 5(2)/5 = 2$, and $b = ar/k = 30/3 = 10$, so D is wrong.
14. **B.** Cell means **are** the means, so set them equal, not to zero.
15. **B.** Equal slopes is exactly no interaction.
16. **B.** Weaker assumptions, never more power, and it does not rescue dependence.
17. **B.** The -1 coding forces deviations to sum to zero, landing the intercept on the grand mean.
18. **B.** His slide says round **down** to the nearest integer.
19. **C.** One equal sign in the system of tests, so 1 df.
20. **B.** Round sample size **up** (12 per group, 48 total). Note the opposite direction to Welch df.
21. **B.** His Unit 1 slide: measurements on the same unit at different times are "repeated measurements", not replicates, and they are not independent.

MY PREDICTION • SIT THIS LIKE THE REAL THING

The 8 questions I expect on your paper

How I built this. He said: 7 to 8 questions, several parts each, comprehensive over Units 1 to 12 and Practice 1 to 9, same style as Test 1 and Test 2, one proof question, 3 hours. I then looked at what he actually set on four of his own papers and what each unit is worth. The **question types** below are a confident prediction. The **numbers** are mine, and his will differ.

Q	TOPIC	CONFIDENCE	EVIDENCE
1	Multiple choice, 4 parts	High	Both his Summer 2026 tests opened this way, 2 marks
2	One-way ANOVA from summaries or R output, plus R^2	Very high	On all four papers, every time
3	Coding: dummy, effect, cell means	Very high	W2023 Q1, W2025 T1 Q4, W2025 T2 Q2, S2026 T2 Q3
4	Contrasts, orthogonality, and a corrected CI	Very high	W2023 Q2, W2025 T2 Q3, S2026 T2 Q4
5	Two-way ANOVA with interaction	Very high	W2023 Q3, S2026 T2 Q5
6	A blocked design: RCB, Latin square or BIBD	High	BIBD was W2023 Q4. Latin square and RCB never tested this term and Practice 7 and 8 cover them
7	ANCOVA, hypotheses in betas	Very high	W2023 Q5 was the biggest question, 8 marks. Practice 9
8	Proof	Certain	He confirmed it on Piazza. One on every paper

THE GAMBLE IN Q6

He has three blocked designs and can only fit one or two. **BIBD** is the one he set on the last final. **Latin square** and **RCB with relative efficiency** have never been tested this term and both have a full practice set. Do all three (C4, E1, E2). It is about 40 minutes and it removes the guess.

Sit it closed-book except the formula sheet. Aim for 2 hours.

P1

[2 MARKS] · MULTIPLE CHOICE

- (a) In a randomized complete block design, the F test on blocks is not significant. This means:
 (A) the treatment test is invalid (B) blocking removed little variability, so a completely randomized design would have been about as good (C) the blocks should be re-randomized (D) the treatments do not differ
- (b) Which is the most serious violation of the ANOVA assumptions?
 (A) normality (B) equal variance (C) independence (D) all equally serious
- (c) In effect coding, the intercept estimates:
 (A) the reference group mean (B) the grand mean (C) zero (D) the largest group mean
- (d) Welch's degrees of freedom comes out as 12.83. You should use:
 (A) 13 (B) 12 (C) 12.83 (D) 14

WHAT THIS QUESTION IS REALLY ASKING

Four quick conceptual checks. None needs a calculator.

SOLUTION P1

- (a) **B.** The block F is a diagnostic about whether the design choice paid off, never a research conclusion. Confirm properly with the relative efficiency.
- (b) **C.** Independence. No adjustment rescues it and the course offers no fix. Normality is the least serious because t and F are robust to it except in small samples.
- (c) **B.** The grand mean. The -1 coding forces the deviations to sum to zero, which is what lands the intercept on the overall average rather than on one group.
- (d) **B.** Round **down** to 12. His Unit 2 slide says so explicitly. Note a required sample size rounds **up**, the opposite direction.

P2

[6 MARKS] · ONE-WAY ANOVA AND EFFECT SIZE

A researcher compares four fertilisers on tomato yield (kg per plot). Plots randomly assigned.

FERTILISER J	N_j	$\bar{Y}_j$	S_j
1	6	18.2	2.1
2	5	21.6	1.8
3	6	17.4	2.4
4	5	22.8	2.0

- (a) Find $\bar{Y}_T$. (b) Find SSA and SSE. (c) Complete the ANOVA table. (d) Test at 5%. (e) Find R^2 and classify it by Cohen's guidelines.

WHAT THIS QUESTION IS REALLY ASKING

One-way ANOVA with both of his favourite traps at once: unbalanced groups and standard deviations rather than variances.

SOLUTION P2

$a = 4$, $N = 6 + 5 + 6 + 5 = 22$. **Unbalanced**, so weight by n_j :

$$\bar{Y}_T = \frac{6(18.2) + 5(21.6) + 6(17.4) + 5(22.8)}{22} = \frac{109.2 + 108 + 104.4 + 114}{22} = \frac{435.6}{22} = 19.8$$

WHY The plain average would be 20.0, which is wrong. Always check the n_j column first.

$$\begin{aligned} SSA &= 6(18.2 - 19.8)^2 + 5(21.6 - 19.8)^2 + 6(17.4 - 19.8)^2 + 5(22.8 - 19.8)^2 \\ &= 6(2.56) + 5(3.24) + 6(5.76) + 5(9) = 15.36 + 16.20 + 34.56 + 45.00 = 111.12 \\ SSE &= 5(2.1^2) + 4(1.8^2) + 5(2.4^2) + 4(2.0^2) = 5(4.41) + 4(3.24) + 5(5.76) + 4(4) \\ &= 22.05 + 12.96 + 28.80 + 16.00 = 79.81 \end{aligned}$$

WHY SSE multipliers are $n_j - 1$, so 5, 4, 5, 4, not all the same. They gave **standard deviations**, so square them.

SOURCE	DF	SS	MS	F
Treatment	3	111.12	37.04	8.3521
Error	18	79.81	4.4339	
Total	21	190.93		

df check: $3 + 18 = 21 = N - 1$ ✓

$$MSA = \frac{111.12}{3} = 37.04, \quad MSE = \frac{79.81}{18} = 4.4339, \quad F = \frac{37.04}{4.4339} = 8.3521$$

WHY Error df = $N - a = 22 - 4 = 18$.

(d) Let μ_j be the mean yield under fertiliser j. $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ vs at least two differ. $F_{0.05,3,18} = 3.16$. Since $8.3521 > 3.16$, **reject**: mean yield differs among the four fertilisers.

WHY Numerator 3, denominator 18.

(e)

$$R^2 = \frac{111.12}{190.93} = 0.5820$$

Cohen's large threshold is $R^2 \approx 0.15$, so this is a **large** effect. About 58.20% of the total variability in yield is explained by which fertiliser the plot received.

WHY Give both the number and the sentence. If the question bans the words low/medium/large, give only the sentence.

P3

[5 MARKS] · CODING

Continuation of P2. Three models, with fertiliser 1 as the dummy reference and fertiliser 4 coded -1 in effect coding:

Dummy: $E[Y] = \beta_0 + \beta_1 I_2 + \beta_2 I_3 + \beta_3 I_4$

Effect: $E[Y] = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3$

Cell means: $E[Y] = \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 I_3 + \gamma_4 I_4$

(a) Estimate β_2 . (b) Estimate α_0 and α_1 . (c) Estimate γ_3 . (d) Write H_0 of equal means in the dummy model and in the cell means model. (e) Express $H_0 : \beta_3 = 0$ in terms of population means.

WHAT THIS QUESTION IS REALLY ASKING

Coding, continuing the same data. Because the design is unbalanced, the effect-coding intercept is the **weighted** grand mean.

SOLUTION P3

CAREFUL: THIS DESIGN IS UNBALANCED

So the grand mean for effect coding is the **weighted** one, $\bar{Y}_T = 19.8$ from P2(a), not the plain average 20.0.

(a) Map the subscripts: $\beta_1 \rightarrow I_2, \beta_2 \rightarrow I_3, \beta_3 \rightarrow I_4$.

$$\hat{\beta}_2 = \bar{Y}_3 - \bar{Y}_1 = 17.4 - 18.2 = \boxed{-0.8}$$

WHY Dummy coefficients are differences from the reference, which is fertiliser 1.

(b)

$$\hat{\alpha}_0 = \bar{Y}_T = \boxed{19.8}, \quad \hat{\alpha}_1 = \bar{Y}_1 - \bar{Y}_T = 18.2 - 19.8 = \boxed{-1.6}$$

WHY Effect coding: intercept is the grand mean, each slope is that group's deviation from it. X_1 flags fertiliser 1.

(c) $\hat{\gamma}_3 = \bar{Y}_3 = \boxed{17.4}$

WHY Cell means coefficients **are** the group means. No intercept, no subtraction.

(d) Dummy: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$. Cell means: $H_0 : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4$.

(e) $\beta_3 = \mu_4 - \mu_1$, so $H_0 : \mu_4 = \mu_1$.

P4

[5 MARKS] · CONTRASTS AND A CORRECTED INTERVAL

Continuation of P2 ($MSE = 4.4339$, $N - a = 18$). The researcher planned these in advance:

$$\psi_1 = \mu_1 - \mu_3, \quad \psi_2 = \mu_2 - \mu_4, \quad \psi_3 = \mu_1 + \mu_3 - \mu_2 - \mu_4$$

(a) Are ψ_1 and ψ_3 orthogonal? (b) Test $H_0 : \psi_3 = 0$ at 5%. (c) Using Bonferroni over all three contrasts, give a 94% CI for ψ_3 . (d) Which procedure would you have used instead if the contrasts had been chosen after seeing the data?

WHAT THIS QUESTION IS REALLY ASKING

Contrasts where the unbalanced design forces the full orthogonality condition with the n values in it, and a Bonferroni interval at an unusual confidence level.

SOLUTION P4

(a) **Unbalanced**, so use the full condition with n_j . $\mathbf{c}_1 = (1, 0, -1, 0)$, $\mathbf{c}_3 = (1, -1, 1, -1)$:

$$\sum_j \frac{c_{1j}c_{3j}}{n_j} = \frac{(1)(1)}{6} + \frac{(0)(-1)}{5} + \frac{(-1)(1)}{6} + \frac{(0)(-1)}{5} = \frac{1}{6} - \frac{1}{6} = 0$$

Orthogonal.

WHY Here the plain dot product also gives 0, but you must **show the divided version** because the design is unbalanced. Marks are for using the right condition.

(b)

$$\hat{\psi}_3 = 18.2 + 17.4 - 21.6 - 22.8 = -8.8$$

$$\sum \frac{c_j^2}{n_j} = \frac{1}{6} + \frac{1}{5} + \frac{1}{6} + \frac{1}{5} = 0.1667 + 0.2 + 0.1667 + 0.2 = 0.7333$$

$$F = \frac{(-8.8)^2}{4.4339(0.7333)} = \frac{77.44}{3.2515} = \boxed{23.8168}$$

$F_{0.05,1,18} = 4.41$, and $23.8168 > 4.41$, so **reject**: fertilisers 1 and 3 together differ from 2 and 4 together.

WHY A contrast always has numerator df 1. Note each $c_j^2 = 1$ here, so the sum is just $\sum 1/n_j$.

(c) 94% confidence $\rightarrow \alpha_{FW} = 0.06$, and $k = 3$:

$$\alpha_B = \frac{0.06}{3} = 0.02 \Rightarrow t_{0.01,18} = 2.552$$

$$s_{\hat{\psi}_3} = \sqrt{4.4339(0.7333)} = \sqrt{3.2515} = 1.8032$$

$$-8.8 \pm 2.552(1.8032) = -8.8 \pm 4.6018 = \boxed{(-13.4018, -4.1982)}$$

WHY Divide alpha by k , then halve again for the two-sided t lookup: $0.02/2 = 0.01$. Zero is outside, agreeing with (b).

(d) **Scheffe**, because Bonferroni and Sidak require the family to be fixed in advance and Tukey only covers pairwise differences. Scheffe is the only valid procedure for contrasts chosen after looking at the data, at the cost of being the most conservative.

P5

[7 MARKS] • TWO-WAY ANOVA

Two factors on battery life (hours). Factor A is material (3 levels), Factor B is temperature (2 levels), with $n = 3$ batteries per cell.

SOURCE	DF	SS	MS	F
Material (A)	?	10684	?	?
Temperature (B)	?	?	39119	?
Interaction	?	9614	?	?
Error	?	18231	?	
Total	?	?		

(a) Complete the table. (b) Test the interaction at 5%. (c) Explain in plain English what an interaction would mean here, without using the word. (d) Test the main effects, if it is legitimate to do so. (e) The cell mean for material 3 at temperature 2 is 108.3, the material 3 row mean is 125.1, the temperature 2 column mean is 92.4, and the grand mean is 105.5. Find $(\alpha\beta)_{32}$.

WHAT THIS QUESTION IS REALLY ASKING

A two-way table with mixed givens and a close interaction call. State the comparison explicitly.

SOLUTION P5

$$a = 3, b = 2, n = 3, \text{ so } N = abn = 18.$$

$$\text{df: } A = 2, B = 1, AB = (3 - 1)(2 - 1) = 2, \text{ Error} = N - ab = 18 - 6 = 12, \text{ Total} = 17. \text{ Check } 2 + 1 + 2 + 12 = 17 \checkmark$$

WHY Get the whole df column from the words before touching any SS. Error df is $N - ab$.

B was given as an **MS**, so $SS_B = 39119 \times 1 = 39119$.

$$MSA = \frac{10684}{2} = 5342, \quad MSAB = \frac{9614}{2} = 4807, \quad MSE = \frac{18231}{12} = 1519.25$$

$$SST = 10684 + 39119 + 9614 + 18231 = 77648$$

WHY Mixing given SS and given MS in one table is exactly what he did on the W2023 final. Read the column headings carefully.

SOURCE	DF	SS	MS	F
Material (A)	2	10684	5342	3.5162
Temperature (B)	1	39119	39119	25.7491
Interaction	2	9614	4807	3.1641
Error	12	18231	1519.25	
Total	17	77648		

(b) $H_0 : (\alpha\beta)_{jk} = 0$ for all j, k , vs at least one non-zero. $F = 3.1641$ vs $F_{0.05,2,12} = 3.89$. Since $3.1641 < 3.89$, **FTR**: no evidence of interaction.

WHY Interaction always first. It is close, so state the comparison explicitly.

(c) It would mean the effect of **material** on battery life is **not the same at every temperature**: the material that lasts longest at one temperature would not be the one that lasts longest at the other, so you could not recommend a material without saying which temperature you meant.

(d) Legitimate, because the interaction was not significant.

Material: $H_0: \alpha_1 = \alpha_2 = \alpha_3 = 0$; $F = 3.5162$ vs $F_{0.05,2,12} = 3.89 \rightarrow \mathbf{FTR}$.

Temperature: $H_0: \beta_1 = \beta_2 = 0$; $F = 25.7491$ vs $F_{0.05,1,12} = 4.75 \rightarrow \mathbf{reject}$, temperature affects battery life.

WHY Different numerator df for the two tests, same denominator 12.

(e)

$$\widehat{(\alpha\beta)}_{32} = \bar{Y}_{32} - \bar{Y}_{A3} - \bar{Y}_{B2} + \bar{Y}_T = 108.3 - 125.1 - 92.4 + 105.5 = \boxed{-3.7}$$

WHY Cell minus row minus column plus grand: what the cell did, minus what pure addition predicted.

P6

[6 MARKS] · A BLOCKED DESIGN

Four diets are compared on weight gain. Because litter matters, five litters are used and each litter supplies four animals, one per diet.

(a) Name the design and write the model, defining every term. (b) Give the full df column for $N = 20$. (c) Given $SS_{\text{Treat}} = 182.4$, $SS_{\text{Block}} = 96.5$, $SST = 340.2$, complete the ANOVA table and test the diets at 5%. (d) Given $MS_{\text{Block}} = 24.125$, compute the relative efficiency and interpret it.

WHAT THIS QUESTION IS REALLY ASKING

A blocked design end to end, finishing with the question that really matters: did blocking pay for itself?

SOLUTION P6

(a) **A randomized complete block design:** litter is a nuisance factor, every diet appears once in every litter, so the blocks are complete.

$$Y_{jk} = \mu + \alpha_j + \beta_k + \varepsilon_{jk}$$

μ overall mean; α_j effect of diet j ; β_k effect of litter k ; $\varepsilon_{jk} \stackrel{iid}{\sim} N(0, \sigma^2)$.

WHY "Each litter supplies one animal per diet" is the definition of complete blocks. Note this is the two-way **additive** model, with no interaction term.

(b) $a = 4$, $b = 5$, $N = 20$. Treatment 3, Block 4, Error $N - a - b + 1 = 20 - 4 - 5 + 1 = 12$, Total 19. Check $3 + 4 + 12 = 19 \checkmark$

WHY Error df also equals $(a - 1)(b - 1) = 3 \times 4 = 12$.

(c) $SSE = 340.2 - 182.4 - 96.5 = 61.3$.

$$MS_{\text{Treat}} = \frac{182.4}{3} = 60.8, \quad MSE = \frac{61.3}{12} = 5.1083, \quad F = \frac{60.8}{5.1083} = 11.9022$$

$H_0: \alpha_1 = \dots = \alpha_4 = 0$. $F_{0.05,3,12} = 3.49$, and $11.9022 > 3.49$, so **reject**: mean weight gain differs among the diets.

WHY SSE is the leftover after both named rows.

(d) $df_{RCB} = 12$, $df_{CR} = N - a = 16$.

$$\begin{aligned} \widehat{RE} &= \frac{(12+1)(16+3)}{(12+3)(16+1)} \cdot \frac{(b-1)MS_B + b(a-1)MSE}{(ab-1)MSE} \\ &= \frac{13(19)}{15(17)} \cdot \frac{4(24.125) + 5(3)(5.1083)}{19(5.1083)} = \frac{247}{255} \cdot \frac{96.5 + 76.6245}{97.0577} \\ &= 0.9686 \times 1.7838 = \boxed{1.7278} \end{aligned}$$

WHY $RE = 1.73 > 1$, so blocking on litter was **worthwhile**: a completely randomized design would have needed about 1.73 times as many animals to estimate the diet differences with the same precision.

P7

[6 MARKS] · ANCOVA

Three teaching programs (Standard, Enhanced, Intensive) on final exam score Y , controlling for prior GPA x . Standard is the reference, and β_1 is the coefficient of x .

(a) Write the parallel planes model with dummy coding, defining every term, and give the table of expected responses. (b) Write H_0 for: (i) controlling for GPA, all three programs have the same mean; (ii) controlling for GPA, Enhanced and Intensive together equal Standard; (iii) GPA has no effect. (c) Write the model allowing the GPA effect to differ by program. (d) Write H_0 for: (i) equal slopes; (ii) controlling for GPA, program has no effect at all. (e) State the extra assumption the parallel planes model needs beyond the usual ANOVA assumptions.

WHAT THIS QUESTION IS REALLY ASKING

The ANCOVA question. All hypotheses, no arithmetic. Build the table, read them off.

SOLUTION P7

(a) Let $I_E = 1$ if the student is in Enhanced and 0 otherwise; $I_I = 1$ if in Intensive and 0 otherwise. Standard is the reference.

$$E[Y|X = x] = \beta_0 + \beta_1 x + \beta_2 I_E + \beta_3 I_I$$

WHY Two indicators for three groups. Defining them in words is worth a mark.

PROGRAM	I_E	I_I	$E[Y X = X]$
Standard (ref)	0	0	$\beta_0 + \beta_1 x$
Enhanced	1	0	$(\beta_0 + \beta_2) + \beta_1 x$
Intensive	0	1	$(\beta_0 + \beta_3) + \beta_1 x$

(b)(i) $H_0 : \beta_2 = \beta_3 = 0$.

(ii) Average the Enhanced and Intensive rows and set equal to Standard:

$$\frac{(\beta_0 + \beta_2) + (\beta_0 + \beta_3)}{2} = \beta_0 \Rightarrow \frac{\beta_2 + \beta_3}{2} = 0 \Rightarrow \boxed{H_0 : \beta_2 + \beta_3 = 0}$$

WHY Different from part (i). Here the two could be non-zero as long as they cancel. Derive it from the table rather than recalling it.

(iii) $H_0 : \beta_1 = 0$.

(c)

$$E[Y|X = x] = \beta_0 + \beta_1 x + \beta_2 I_E + \beta_3 I_I + \beta_4 x I_E + \beta_5 x I_I$$

slopes: Standard β_1 , Enhanced $\beta_1 + \beta_4$, Intensive $\beta_1 + \beta_5$.

WHY One interaction term per non-reference group.

(d)(i) $H_0 : \beta_4 = \beta_5 = 0$. (ii) $H_0 : \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$.

(e) **Homogeneity of regression slopes:** the relationship between Y and the covariate must have the same slope in every program, that is, no interaction between program and covariate. Also the relationship between Y and x must be **linear**.

P8

[5 MARKS] · THE PROOF

One-way model $Y_{ij} = \mu_j + \varepsilon_{ij}$, $\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$, $i = 1, \dots, n_j$, $j = 1, \dots, a$. Let $\hat{\psi} = \sum_j c_j \bar{Y}_j$ with $\sum_j c_j = 0$.

(a) Derive $V(\bar{Y}_j)$. (b) Derive $V(\hat{\psi})$. (c) Hence state the distribution of $\hat{\psi}$ and explain why the test statistic uses MSE rather than σ^2 .

Show all steps and explicitly name the theorem or fact used at each step.

WHAT THIS QUESTION IS REALLY ASKING

A proof that walks from one variance to a distribution, then asks the conceptual question of why MSE replaces sigma squared.

SOLUTION P8

(a)

$$V(\bar{Y}_j) = V\left(\frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ij}\right) \stackrel{\text{by independence}}{=} \frac{1}{n_j^2} \sum_{i=1}^{n_j} V(Y_{ij}) = \frac{n_j \sigma^2}{n_j^2} = \boxed{\frac{\sigma^2}{n_j}}$$

WHY The constant comes out **squared**. Independence is what lets the variance of a sum become the sum of variances, and saying so is worth a mark.

(b)

$$V(\hat{\psi}) = V\left(\sum_{j=1}^a c_j \bar{Y}_j\right) \stackrel{\text{by independence}}{=} \sum_{j=1}^a c_j^2 V(\bar{Y}_j) = \sum_{j=1}^a c_j^2 \frac{\sigma^2}{n_j} = \boxed{\sigma^2 \sum_{j=1}^a \frac{c_j^2}{n_j}}$$

WHY Different treatment groups use disjoint observations, so they are independent and all covariance terms vanish. Each c_j comes out squared.

(c) Each $\bar{Y}_j$ is normal, and a **linear combination of independent normals is normal**, so

$$\hat{\psi} \sim N\left(\sum_j c_j \mu_j, \sigma^2 \sum_j \frac{c_j^2}{n_j}\right)$$

WHY Name the fact. The mean follows from $E[\bar{Y}_j] = \mu_j$ and linearity.

Because σ^2 is **unknown**, we replace it by its unbiased estimator MSE . Standardising with an estimated variance turns the normal into a t :

$$\frac{\hat{\psi} - \psi}{\sqrt{MSE \sum_j c_j^2 / n_j}} \sim t_{N-a}$$

WHY This is the same reason a one-sample z becomes a t . The df is $N - a$, the df of the MSE used in the denominator. Squaring gives $F_{1, N-a}$, which is why either test may be used.

AFTER YOU SIT IT

Mark yourself out of 42. Anything you could not start is a **topic** gap, not a numbers gap: go to the matching lettered question above (P2→A2/C1, P3→B2/D2, P4→B3/C2, P5→C3, P6→E1, P7→C5, P8→B4/D5) and do that one too.

LAST SECTION

Tonight and tomorrow

TONIGHT, IN THIS ORDER

1. **Paper A** (A1 to A5). It is the gentlest and it establishes every symbol. Do it with a pen, covering the solutions.
2. **Paper C** questions C1, C3, C4, C5. Those are 9, 7, 7 and 8 marks of the real final: one-way plus coding, two-way, BIBD, ANCOVA.
3. **All the proofs:** A5, B4, C6, D5, V7, V8. Write the words **by independence** and **divide by** σ^2 every time.
4. **Paper B** B1 and B3 for post-hoc and orthogonal sets, then **D3** for Sidak.
5. **The gaps:** E1, E2, E4, and variants V1, V2, V13, V16. None of these appeared on the past papers and you have 7 to 8 questions this time.
6. **MC bank.** It is fast and it is 2 marks.

EXAM MORNING

1. EX 200, 255 McCaul, 9:00. Arrive early; latecomers get no extra time. Student card and an approved calculator with working batteries.
2. Once it starts you cannot leave the room for any reason.
3. **Write your name on the formula sheet.** It gets handed in with the booklet.
4. Read page 4 of the sheet, the General Facts block, before writing anything. That is the proof toolkit.
5. Scan all 7 to 8 questions and start with the one whose design you can name instantly.
6. 4 decimals, $\alpha = 0.05$ unless stated, circle final answers, define μ_j in words, never write "accept H_0 ".

LAST FIVE MINUTES

Check every df column adds to $N - 1$. Check every contrast's coefficients sum to zero. Check the Q_i sum to zero. Check every effect estimate set sums to zero. Check every conclusion has a sentence in the words of the problem.

APPROVED CALCULATORS

Casio fx-260 Solar II, fx-300MS Plus, fx-300ES Plus (either edition); TI-30Xa, TI-30XIIS, TI-30XS MultiView, TI-34 MultiView; Sharp EL-531X, EL-531XT, EL-531TG. Colour variants fine. **Not allowed:** TI-36X Pro, all Casio fx-991 and fx-115, every TI-83, TI-84, TI-89, TI-Nspire.

Personal revision notes for STA305H1 S, Summer 2026, Dr. Luai Al Labadi, University of Toronto. Questions quoted from the posted W2025 Term Tests 1 and 2, the W2023 final and its answer key, the Summer 2026 Test 1 and Test 2 solutions, and Practice Problems 1 to 9, for private study. All arithmetic re-derived and checked against the posted keys.